

Elliptic Curves

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Contents

1	Introduction	2
2	Group Law	3
2.1	Exercise	4
3	Singular Weierstrass Equation	4
4	Review of Algebraic Curves	5
5	Geometry of Elliptic Curves	6
5.1	Exercise	8
6	Elliptic Curves over $\mathbb{C}$	9
7	Elliptic Curves over $\mathbb{R}$	10
8	Isogeny	11
9	Elliptic Curves over Finite Fields	12
9.1	Exercise	14
10	Formal Group	15
11	Elliptic Curves over Local Fields	18
12	Elliptic Curves over Number Fields	21
12.1	Exercise	22
13	Kummer 's Theorem	23
14	Weak Mordell-Wéil	24
14.1	Exercise	25
15	Heights	25
15.1	Exercise	27
16	Dual Isogenies and the Wéil Pairing	27
17	Galois Cohomology	29
18	Descent by cyclic isogeny	32
18.1	Descent by 2-isogeny	33
18.2	Exercise	34
19	Afterthoughts	35

These notes are based on the handwritten notes by the lecturer, which essentially covers the main body of AEC by Silverman except for the sections on Tate modules. These proofs of showing that degree map and canonical heights function are quadratic forms are omitted.

1 Introduction

A central goal of number theory is to find solutions of *Diophantine equations*, i.e. find integral/rational solutions to systems of polynomial equations with integral/rational coefficient. This is equivalent to find rational points on algebraic varieties defined over $\mathbb{Q}$

Example 1. Consider the circle $C : x^2 + y^2 = 1$ and line $L : y = t(x + 1)$. Their second intersection P has coordinate in $\mathbb{Q}$ if and only if $t \in \mathbb{Q}$. More explicitly, we can give the coordinates of P in terms of rational polynomials in t with $P = P_t = (\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2})$. This gives a *parameterization* of rational points on C

Now the rational map $\mathbb{A}^1 \rightarrow C, ; t \mapsto (\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2})$ can extend to an isomorphism of smooth projective curves

$$\mathbb{P}^1 \rightarrow \widehat{C} = V(x^2 + y^2 - z^2) \subset \mathbb{P}^2; [u : v] \mapsto [v^2 - u^2 : 2uv : v^2 + u^2]$$

over $\mathbb{Q}$. This induces a bijection on rational points

$$\mathbb{Q} \cup \{\infty\} = \mathbb{P}^1(\mathbb{Q}) \leftrightarrow \widehat{C}(\mathbb{Q}) = C(\mathbb{Q})$$

because $V(z) \cap V(x^2 + y^2 - z^2)$ has no rational points.

More generally, if $X \subset \mathbb{P}^2$ is a smooth *conic* over $\mathbb{Q}$ (i.e. defined by $f = 0$ where $f(x, y, z)$ is a homogeneous polynomial of degree 2). Then same argument gives $X(\mathbb{Q}) = \emptyset$ or $X \cong \mathbb{P}^1$ over $\mathbb{Q}$.

Example 2. (congruent number problem) We say $n \in \mathbb{Z}_{>0}$ is a **congruent number** if there is a triangle with rational side lengths and area n

Proposition 1.1. n congruent if and only if $ny^2 = x^3 - x$ has solution over $\mathbb{Q}$

Proof. n is congruent if and only if there are $a, b, c \in \mathbb{Q}^\times$ satisfying $a^2 + b^2 = c^2$ and $ab = 2n$. By replacing a, b with $a/c, b/c$ and example 1, we can parametrize them with $t \in \mathbb{Q}$ such that $(a/c, b/c) = (\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2})$. Then we can compute that $n/c^2 = (1-t^2)t/(1+t^2)^2$. Then we set $y = (1+t^2)/c$ and $x = -t$, and recover $ny^2 = x^3 - x$ $\square$

Remark 1.1. Note $y \neq 0$ and so the resulting point is not 2-torsion.

Remark 1.2. Change of variable $x' = nx, y' = n^2y$ transforms $ny^2 = x^3 - x$ into

$$y'^2 = x'^3 - n^2x'$$

This is an example of elliptic curves.

Definition 1.1. For a field K , a **Weierstrass equation** over K is of the form

$$y^2z + a_1xyz + a_3yz^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3$$

To ease notations identify it with

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$$

Remark 1.3. The first equation defines an irreducible plane cubic $E \subset \mathbb{P}^2$, while the second is an affine patch of $D_+(z) \cong \text{Spec} K[x/z, y/z]$

The **points at infinity** is $\mathcal{O}_E = [0 : 1 : 0] \in E$ by setting $z = 0$ in the equation. Let $E(K) = \{\mathcal{O}_E\} \cup$ K -points satisfying the second equation.

Definition 1.2. An elliptic curve E/K is a smooth curve given by a Weierstrass equation over K .

Remark 1.4. When $\text{char}(K) \neq 2$, change of variables simplify the second equation into $y^2 = f(x)$ where $f(x)$ is a monic cubic.

When $\text{char}(K) \neq 2, 3$, it can be further simplified to $y^2 = x^3 + ax + b$

For any K , a Weierstrass equation defines an elliptic curve if and only if the *discriminant* $\Delta \neq 0$. For the most simplified $y^2 = x^3 + ax + b$, we have $\Delta = -16(4a^3 + 27b^2)$

For this course we will study $E(K)$ for elliptic curves E/K when K is $\mathbb{C}$, finite fields, local fields and number fields.

Moral: $E(K)$ is naturally an abelian group!

(Fermat's result shows that for $E : y^2 = x^3 - x$, $E(\mathbb{Q}) = \{(0, 0), (\pm 1, 0), \mathcal{O}_E\} \cong (\mathbb{Z}/2\mathbb{Z})^2$, which is an important fact to study elliptic curves of similar forms)

In this course, K will always be assumed perfect.

Proposition 1.2. Suppose $\text{char}(K) \neq 2$, $E : y^2 = f(x)$ an elliptic curve over K . Suppose $x(t), y(t)$ satisfying $y(t)^2 = f(x(t))$. Then $x(t), y(t) \in K$. That is, $E(K) = E(K(t))$

Proof. Assume $\overline{K} = K$, $E : y^2 = x(x-1)(x-\lambda)$ for $\lambda \in K \setminus \{0, 1\}$ in Legendre form. Suppose $P = (x(t), y(t)) \in E(K(t))$. Write $x(t) = u/v$ with $u, v \in K[t]$ coprime. Define $h(P) = \max\{\deg(u), \deg(v)\}$.

WLOG, assume $h(P)$ is minimal. Then we have $w^2 = uv(u-v)(u-\lambda v)$ with $w = yv^2 \in K[t]$. Since $u, v, u-v, u-\lambda v$ are coprime, there are $a, b, c, d \in K[t]$ such that $u = a^2$ and so on. Hence $(cd)^2 = (u-v)(u-\lambda v) = (a-b)(a+b)(a-\mu b)(a+\mu b)$ for some $\mu \in K$ with $\mu^2 = \lambda$ since K is perfect and $\text{char}(K) \neq 2$.

Now we set $\alpha = (1-\mu)(a+b)/2$, $\beta = (1+\mu)(b-a)/2$. We can verify that $\gamma^2 = \alpha\beta(\alpha-\beta)(\alpha-\lambda'\beta)$ with $\lambda' = (1-\mu/1+\mu)^2$ and $\gamma = \sqrt{-1}cd(1-\mu)/2$. Then $P = (\alpha/\beta, \gamma/\beta^2) \in E'(K(t))$ where E' is in the Legendre form by λ' . Since $u = a^2$, $v = b^2$, we have $h(P') = \frac{1}{2}h(P)$

Repeat once again, setting $\mu' = (\frac{1-\mu}{1+\mu})^2$. We compute that $(\frac{1-\mu'}{1+\mu'})^2 = \lambda$. So we get back to E/K and $P'' \in E(K(t))$ with $h(P'') = \frac{1}{4}h(P)$, contradiction. $\square$

2 Group Law

Given a field K , E/K an elliptic curve given by a Weierstrass equation. Since E is a smooth plane cubic, each line in $\mathbb{P}^2$ meets E in 3 points with multiplicities by Bezout's theorem.

Definition 2.1. (*Chord-tangent Progress*) Given $P, Q \in E(K)$, $P \oplus Q$ is defined as follows: let S be the third point of intersection of E and the line PQ , or tangent line at P if $P = Q$. Then $P \oplus Q$ is defined as the third intersection of point of E and the line SO_E .

Remark 2.1. If $P \in E(K) \setminus \{\mathcal{O}_E\}$ then $\mathcal{O}_E P$ is the vertical line through P

Remark 2.2. Tangent at $\mathcal{O}_E$ is $V(z) \subset \mathbb{P}^2$ and meets E at $\mathcal{O}_E$ with multiplicity 3.

Theorem 2.1. $(E(K), \oplus)$ is an abelian group.

Proof. It is clearly commutative. The identity is $\mathcal{O}_E$ and the inverse of P is the third intersection of the line $\mathcal{O}_E P$ and E . The associativity has many ways to prove, and we are about to see. $\square$

Remark 2.3. $P \oplus Q \oplus R = \mathcal{O}_E$ if and only if P, Q, R are colinear.

(Explicit formula for group law) Given $E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$, for $P = (x, y) \in E(K)$, we have $-P = (x, -a_1x - y)$. Now for $P_1 = (x_1, y_1), P_2 = (x_2, y_2) \in E(K)$, write $y = \lambda x + v$ for the line P_1P_2 . Substitute this into the equation for E , we get $x^3 + (a_2 - \lambda^2 - a_1\lambda)x^2 + \dots = 0$. Let $P_1 \oplus P_2 = (x_3, y_3)$. Then we get

$$x_3 = \lambda^2 + a_1\lambda - a_2 - x_1 - x_2, \quad y_3 = -(\lambda + a_1)x_3 - v - a_3$$

Now we need to find λ, v . If $P_1 \neq P_2$ then $\lambda = \frac{y_2 - y_1}{x_2 - x_1}$ and $v = \frac{y_1x_2 - y_2x_1}{x_2 - x_1}$. If $P_1 = P_2$, we need to find the equation of its tangent line, which can be done by taking partial derivation.

2.1 Exercise

In elliptic curves some explicit computation seems inevitable. We start with some easy one as warm-up

Example 3. Let $E/\mathbb{Q} : y^2 - y = x^3 - x^2$. We want to verify that $P = (0, 0)$ has order 5. Now $-P = (0, 1)$ and the tangent at P is $y = 0$. It intersects with E at P and $(1, 0)$. So $2P = -(1, 0) = (1, 1)$. The tangent at $2P$ is $y = x$. It intersects with E with $2P$ and $(0, 0) = P$. So $4P + P = 0$ as required.

We now give the formula for doubling a point on $E : y^2 = x^3 + ax + b$. Let $P = (x, y)$ with $y \neq 0$, since $2P = 0$ if and only if $y = 0$ in this case. The tangent of P has slope $\frac{3x^2 + a}{2y}$. Then after a long computations eventually we get $2P = (\frac{x^4 - 2ax^2 - 8bx + a^2}{4y^2} = \frac{f'(x)^2 - 8xf(x)}{4f(x)}, \frac{x^6 + 5ax^4 + 20bx^3 - 5a^2x^2 - 4abx - a^3 - 8b^2}{8y^3} = \frac{f'(x)^3 - 12xf(x)f'(x) + 8f(x)^2}{8y^3})$. Of course it is unpractical to memorize this and plug in the coordinates when we need to. I think it is enough to know how to compute the tangent. Using this, we can compute the 3-torsion points on E , because $3T = 0 \implies 2T = -T$ and so $x = \frac{f'(x)^2 - 8xf(x)}{4f(x)} \implies 3x^4 + 6ax^2 + 12bx - a^2 = 0$.

3 Singular Weierstrass Equation

Although we are most interested in elliptic curves in this course, it is worth noting that the study of the curves given by singular Weierstrass equations is very complete and easier to get into.

Given K a perfect field, $\text{char}(K) \neq 2$ and $E : y^2 = f(x)$ defined over K with $f(x)$ a monic cubic. Assume E is singular, i.e. $f(x)$ has repeated roots. In this E has a *unique* singular point $(r, 0)$ (because the multiplicity at a singular point is at least 2) for r a repeated root of $f(x)$.

Since Galois action maps singular points to singular points, r is fixed by Γ_K and hence $r \in K$. Also, we can translate the singular point to $(0, 0)$ and get two types

1. $E : y^2 = x^3$ when f has a triple root. In this case $(0, 0)$ is a **cusp**
2. $E : y^2 = x^2(x + \lambda)$ when f has a double root. In this case $(0, 0)$ is a **node**, as there are two tangent lines to the curve at $(0, 0)$

By Bezout's theorem, any line passing through $(0, 0)$ meets E at a unique third point (possibly be $(0, 0)$). As in example 1, we can therefore parametrize E

1. For type 1, $\mathbb{A}^1 \rightarrow E, t \mapsto (t^2, t^3)$

2. For type 2, $\mathbb{A}^1 \rightarrow E, t \mapsto (t^2 - \lambda, t(t^2 - \lambda))$

In each case, inverse rational map is $(x, y) \mapsto y/x$ In fact,

Proposition 3.1. 1. *Chord-tangent process makes $E_{ns}(K) = E(K) \setminus \{\text{singular point}\}$ into an abelian group*

2. *In type 1 we have an isomorphism*

$$\phi : E_{ns}(K) \xrightarrow{\sim} (K, +)$$

$$(x, y) \mapsto x/y, \mathcal{O}_E \mapsto 0$$

3. *In type 2, we have an isomorphism*

$$\phi : E_{ns}(\overline{K}) \xrightarrow{\sim} (\overline{K}^\times, \times)$$

$$(x, y) \mapsto \frac{y + \sqrt{\lambda}x}{y - \sqrt{\lambda}x}, \mathcal{O}_E \mapsto 1$$

which restricts to $E_{ns}(K) = K$ if $\sqrt{\lambda} \in K$, $E_{ns}(K) = \ker(K(\sqrt{\lambda}) \xrightarrow{\text{norm}} K)$ if $\sqrt{\lambda} \notin K$

For (2), we recover its homogeneous form $y^2z = x^3$ and look at the affine chart $\{y \neq 0\}$. Then E_{ns} is $\{u = v^3\} \subset \mathbb{A}^2$ and $\mathcal{O}_E = 0$. Then ϕ becomes $\phi : (u, v) \mapsto v$, which is clearly bijective. We can also prove that it is a homomorphism without much effort. For (3), see the corresponding section in *AEC*.

4 Review of Algebraic Curves

In this section, $K = \overline{K}$, C is a smooth projective curve over K . $K(C)$ is the **field of rational functions** on C .

Definition 4.1. A **divisor** on C is a formal sum $D = \sum_{P \in C} n_P(P)$. It is **effective** if $n_P \geq 0$ for all P and we write $D \geq 0$. We write $\text{Div}(C)$ the abelian group of divisors on C . We have the degree map $\text{deg} : \text{Div}(C) \rightarrow \mathbb{Z}, \sum_{P \in C} n_P(P) \mapsto \sum n_P$. We write $\ker(\text{deg}) = \text{Div}^0(C)$

For $P \in C$ we have the local ring $\mathcal{O}_P$ the rational functions regular at P with maximal ideal m_P the rational functions that vanish at P . Moreover, $\mathcal{O}_P$ is a DVR with associated discrete valuation ord_P on $K(C)^\times$. $t_P \in K(C)^\times$ is a **uniformizer** at P if $\text{ord}_P(t_P) = 1$

For $f \in K(C)^\times$, define $\text{div}(f) = \sum_{P \in C} \text{ord}_P(f)(P) \in \text{Div}^0(C)$ (pullback of $(0) - (\infty)$). We call divisors of this form **principal**. Two divisors D_1, D_2 are **linearly equivalent** if $D_1 - D_2$ is principal, and we write $D_1 \sim D_2$. The **class group** is $\text{Cl}(C) = \text{Div}(C) / \sim$ and $\text{Cl}^0(C) = \text{Div}^0(C) / \sim$

Example 4. Suppose $\text{char}(K) \neq 2$. Consider elliptic curve $y^2 = (x - e_1)(x - e_2)(x - e_3)$ no multiple root. Let $P = (x(P), y(P)) \in E$. Then $x - x(P)$ is a uniformizer at P if $x(P) \notin \{e_1, e_2, e_3\}$. If $x(P) \in \{e_1, e_2, e_3\}$, then y is a uniformizer at P and $\text{ord}_P(x - x(P)) = 2$ which can be seen from the equations. This shows that $\text{div}(y) = (e_1, 0) + (e_2, 0) + (e_3, 0) - 3\mathcal{O}_E$ and $\text{div}(x) = (0, \sqrt{f(0)}) + (0, -\sqrt{f(0)}) - 2\mathcal{O}_E$, and hence x/y is a uniformizer at $\mathcal{O}_E$

Definition 4.2. The space of meromorphic (pr rational) differentials on C , Ω_C is the $K(C)$ -vector space spanned by symbols $\{df \mid f \in K(C)\}$ subject to the relations of normal derivations, e.g. $d(ab) = a \cdot d(b) + d(a) \cdot b$

Facts: Ω_C is a 1-dimensional vector space over $K(C)$.

Definition 4.3. Let $0 \neq \omega \in \Omega_C$, t_P the uniformizer at P . Then $\omega = f dt_P$ for some $f \in K(C)^\times$. Define $\text{ord}_P(\omega) = \text{ord}_P(f)$, and $\text{div}(\omega) = \sum \text{ord}_P(\omega)(P) \in \text{Div}(C)$. The **canonical divisor** $K_C = [\text{div}(\omega)] \in \text{Cl}(C)$ for any $0 \neq \omega \in \Omega_C$. This is well-defined because any two nonzero differentials differ by some $f \in K(C)^\times$ since Ω_C is of dimension 1 over $K(C)$.

Definition 4.4. Let $D \in \text{Div}(C)$. Let $\mathcal{L}(D) = \{f \in K(C) \mid \text{div}(f) + D \geq 0\} \cup \{0\}$. This is a finite dimensional K -vector space and $D_1 \sim D_2$ implies $\mathcal{L}(D_1) \cong \mathcal{L}(D_2)$

Definition 4.5. The **genus** of C is $g(C) = \dim_K(\mathcal{L}(K_C)) = \dim_K(\{\omega \in \Omega_C \mid \text{div}(\omega) \geq 0\})$, i.e. the K -vector space of regular differentials

Theorem 4.1. (Riemann-Roch) Let $D \in \text{Div}(C)$, then $\dim_K(\mathcal{L}(D)) - \dim_K(\mathcal{L}(K_C - D)) = \deg D + 1 - g$.

Remark 4.1. Take $D = K_C$, we get $\deg(K_C) = 2g - 2$

Example 5. With the settings in example 3, let $\omega = dx/y$. We claim that $\text{div}(\omega) = 0$

For $P = (x(P), y(P))$ with $x \notin \{e_1, e_2, e_3\}$, then $\omega = d(x - x(P))/y$ and $\text{ord}_P(\omega) = \text{ord}_P(y) = 0$. For $P = (e_i, 0)$, $y^2 = f(x) \implies 2ydy = f'(x)dx$. So $\omega = 2dy/f'(x)$, and $\text{ord}_P(\omega) = 0$ since $f(x)$ has no multiple roots. Moreover, $\omega = 2f(x)d(x/y)/(2f(x) - xf'(x))$, so $\text{ord}_{\mathcal{O}_E}(\omega) = 0$

So $K_E = 0$ and $g = 1$

Remark 4.2. For elliptic curves given by Weierstrass equations, $\text{div}(\omega) = 0$ for $\omega = dx/(2y + a_1x + a_3)$. This is the **invariant differential**

Facts: for X a projective variety, $\phi : C \dashrightarrow X$ rational map. Then ϕ is regular, i.e. a morphism. Moreover, for X, Y smooth projective curves and $\phi : X \rightarrow Y$ non-constant morphism, ϕ is surjective and $\phi^* : K(Y) \rightarrow K(X); f \mapsto f \circ \phi$ is injective.

The **degree** of ϕ is $[K(X) : \phi^*K(Y)] < \infty$. ϕ is separable if the field extension is separable. Given $P \in Y$ and $Q \in \phi^{-1}(P)$, the **ramification index** of ϕ at Q is $e_\phi(Q) = \text{ord}_Q(\phi^*t_P)$

Theorem 4.2. For all $P \in Y$, $\deg(\phi) = \sum_{Q \in \phi^{-1}(P)} e_\phi(Q)$

Theorem 4.3. If ϕ is separable, then $\#\phi^{-1}(P) = \deg \phi$ for all but finitely many $P \in Y$

Similarly, ϕ induces pushforward and pullback on the differentials and divisors. We might use $\phi^*\text{div}(f) = \text{div}(\phi^*f)$

Remark 4.3. For $\text{char}(K) = 0$, we have the Riemann-Hurwitz formula, which implies $g(X) \geq g(Y)$ for $\phi : X \rightarrow Y$. So there is no non-constant morphism from $\mathbb{P}^1$ to elliptic curves.

5 Geometry of Elliptic Curves

Proposition 5.1. Given $K = \overline{K}$, E/K elliptic curve. Define $E \rightarrow \text{Cl}^0(E)$ given by $P \mapsto [(P) - (\mathcal{O}_E)]$. Then ϕ is an isomorphism.

Proof. For $D \in \text{Div}(E)$ with $\deg(D) > 0$, Riemann-Roch and $g(E) = 1$ gives $\deg(D) = \dim_K(\mathcal{L}(D))$. Now suppose $\phi(P) = \phi(Q)$. Then $(P) - (\mathcal{O}_E) \sim (Q) - (\mathcal{O}_E) \implies (P) \sim (Q)$. Then $(P) = \text{div}(f) + (Q)$ for some f and so $f \in \mathcal{L}((Q)) = K^\times$ by the former observation. So $\text{div}(f) = 0$ and $P = Q$. So ϕ is injective.

Let $[D] \in \text{Cl}^0(E)$. Then $\deg(\mathcal{O}_E + D) = 1$. So there is $f \in K(C)^\times$ such that $\text{div}(f) + D + \mathcal{O}_E \geq 0$. Then $\deg(\text{div}(f) + D + \mathcal{O}_E) = 1$ implies $\text{div}(f) + D + \mathcal{O}_E = (P)$ for some P . So $D \sim (P) - (\mathcal{O}_E)$ and $\phi(P) = [D]$. So ϕ is surjective.

Now for any $P, Q \in E(K)$, let $L_1 : f(x, y, z) = 0 \subset \mathbb{P}^2$ the line through P, Q , with third point of intersection $R \in E(K)$. Let $L_2 : f'(x, y, z) = 0 \subset \mathbb{P}^2$ be the line through $R, \mathcal{O}_E$, and the third point of intersection is exactly $P \oplus Q$. Then $f/f' \in K(C)^\times$ and $\text{div}(f/f') = (P) + (Q) + (R) - (R) - (P \oplus Q) - (\mathcal{O}_E) \implies (P) + (Q) - 2(\mathcal{O}_E) \sim (P \oplus Q) - (\mathcal{O}_E) \implies \phi(P) + \phi(Q) = \phi(P \oplus Q)$ $\square$

On the other hand, this gives a proof of showing that $(E(K), \oplus)$ has associativity, since $Cl^0(E)$ is a group.

Definition 5.1. An *elliptic curve over K* is a pair $(E, \mathcal{O}_E)$, where E is a smooth projective curve with genus 1 over K and $\mathcal{O}_E \in E(K)$. An *isomorphism of elliptic curves over K* is an isomorphism of curves defined over K that send $\mathcal{O}_E$ to $\mathcal{O}_{E'}$

Theorem 5.1. E/K an elliptic curve in the sense of definition 5.1. Then E is isomorphic to an elliptic curve in the Weierstrass form and $\mathcal{O}_E \mapsto [0 : 1 : 0]$.

Proof. We may assume $K = \bar{K}$. By Riemann-Roch, $\dim_K(\mathcal{L}(n(\mathcal{O}_E))) = n$ for all $n \geq 1$. Then $\mathcal{L}(\mathcal{O}_E) = K$, $\mathcal{L}(2\mathcal{O}_E)$ is the span of 1 and x because $\mathcal{O}_E$ is the only pole of x and $\text{ord}_{\mathcal{O}_E}(x) = -2$. Similarly $\mathcal{L}(3\mathcal{O}_E)$ is the span of 1, x and y .

Finally, we look at $\mathcal{L}(6\mathcal{O}_E)$. $1, x, y, x^2, xy, y^2, x^3$ are all in $\mathcal{L}(6\mathcal{O}_E)$ which is a k -vector space of dimension 6. And removing x^3 or y^2 gives a basis of $\mathcal{L}(6\mathcal{O}_E)$ by looking at their orders of pole at $\mathcal{O}_E$. After rescaling x, y if necessary, we recover $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$. Let E' be the cut-out curve in $\mathbb{P}^2$

Now E is smooth, so the rational map $\phi : E \dashrightarrow E'; P \mapsto [x(P) : y(P) : 1]$ is a morphism. Moreover, $[x(P) : y(P) : 1] = [x/y(P) : 1 : 1/y(P)]$, so $\mathcal{O}_E \mapsto [0 : 1 : 0]$. By the tower law on their function fields, ϕ is a birational isomorphism. If E' is singular, E' is birational to $\mathbb{P}^1$ as in section 3 and hence E isomorphic to $\mathbb{P}^1$, a contradiction since $g(E) = 1$ and $g(\mathbb{P}^1) = 0$.

Thus E' is smooth and the rational map $\phi^{-1} : E' \rightarrow E$ is a morphism. So $E \cong E'$ $\square$

Facts: if $D \in \text{Div}(E)$ is defined over K , i.e. invariant under Γ_K . Then $\mathcal{L}(D)$ has a basis of rational functions defined over K .

Proposition 5.2. $E/K, E'/K$ two elliptic curves in Weierstrass form. Then $E \cong E'$ over K if and only if the Weierstrass equations are related by change of variables $x = u^2x' + r$, $y = u^3y' + u^2sx' + t$ with $u, r, s, t \in K$ and $u \neq 0$

This can be shown by analyzing $\mathcal{L}(2\mathcal{O}_E)$ and $\mathcal{L}(3\mathcal{O}_E)$. In practice we will mostly use the change of variable $x = u^2x'$ and $y = u^3y'$

Corollary 5.1. $E : y^2 = x^3 + ax + b, E' : y^2 = x^3 + a'x + b'$. Then $E \cong E'$ over K if and only if $a = u^4a'$, $b = u^6b'$ for some $u \in K^\times$

Definition 5.2. The *j -invariant* of $E : y^2 = x^3 + ax + b$ is $j(E) = \frac{1728(4a^3)}{4a^3 + 27b^2}$

Corollary 5.2. With the setting s in corollary 5.1, $E \cong E'$ implies if $\Delta(E) = u^{12}\Delta(E')$ and $j(E) = j(E')$. If $K = \bar{K}$, $j(E) = j(E')$ if and only if $E \cong E'$

Theorem 5.2. Elliptic curves are *group varieties*, i.e. the inverse map $i : P \mapsto -P$ and the addition $m : (P, Q) \mapsto P \oplus Q$ are morphisms.

Indeed the explicit formula shows that i and m are just rational polynomials, and hence define rational maps on E and $E \times E$. Since E is smooth, i is in fact a morphism. We need to show

that m is regular on $E \times E$. By the explicit formula we see that m is already regular on $U = E \times E \setminus \{(P, Q) | P, Q, P - Q, P + Q = \mathcal{O}_E\}$.

For $P \in E$ we define $T_P : E \rightarrow E; Q \mapsto Q + P$ which is a morphism. We consider the composition of maps

$$E \times E \xrightarrow{T_{-P} \times T_{-Q}} E \times E \xrightarrow{m} E \xrightarrow{T_{P+Q}} E$$

which has the same effect as m , but regular on $T_P \times T_Q(U)$. As P, Q vary this covers $E \times E$. So m is a morphism

Definition 5.3. For $n \geq 0$ we define the morphism $[n] : E \rightarrow E; P \mapsto nP$ and $[-n] = [-1] \circ [n]$. The **n -torsion subgroup** of E is $E[n] = \ker[n]$

To understand $E[n]$, we start with elliptic curves over $\mathbb{C}$

5.1 Exercise

Next we study the **twists** of elliptic curves. Recall that a scheme X' over K is a twist of X over K if $X_{\overline{K}} \cong X'_{\overline{K}}$

Proposition 5.3. Let $E/\mathbb{Q} : y^2 = f(x)$ in Weierstrass form. Define $E_d : dy^2 = f(x)$ which can be put into Weierstrass form by $y \mapsto y/d^2, x \mapsto x/d$. If $j(E) \neq 0, 1728$, then every twist of E is isomorphic to E_d for some unique square-free integer d .

Proof. We can take the short Weierstrass form $E : y^2 = x^3 + ax + b$. Then E_d is isomorphic to $y^2 = x^3 + d^2ax + d^3b$. Suppose $E'/\mathbb{Q} : y^2 = x^3 + a'x + b'$ is a twist of E , i.e. E is isomorphic to E' over $\overline{\mathbb{Q}}$. By corollary 5.1, there is $u \in \overline{\mathbb{Q}}$ such that $a = u^4a', b = u^6b'$. Since $j(E) = \frac{1728(4a^3)}{4a^3 + 27b^2} \neq 0, 1728$, we see that $a, b \neq 0$. So we can take $d = u^{-2} = ab'/a'b \in \mathbb{Q}$ and then $E_d \cong E'$. So far, d is just a non-zero rational number, but a similar argument to the one above shows that $E_1 : y^2 = x^3 + d^2ax + d^3b$ is isomorphic to $E_2 : y^2 = x^3 + d'^2ax + d'^3b$ over $\mathbb{Q}$ if and only there is $u \in \mathbb{Q}$ such that $u^2 = d/d'$. So we can make each $v_p(d) \in \{0, 1\}$ for each prime p and get a square-free d . $\square$

E and E_d are said to be **quadratic twists**.

To get some more flavor on how the isomorphism of elliptic curves goes, we consider the following proposition:

Proposition 5.4. Let $E/\mathbb{F}_2$ be the elliptic curve $y^2 + y = x^3$. Show that the group $\text{Aut}(E)$ of automorphisms of E defined over $\overline{\mathbb{F}_2}$ is a non-abelian group of order 24

As in proposition 5.2, any automorphism of E necessarily has the form $(x, y) \mapsto (u^2x + r, u^3y + u^2sx + t)$ for $u, s, t, r \in \overline{\mathbb{F}_2}$. In the formula sheet the relations of a_i and a'_i are given, and we plug in $a_1 = a_2 = a_4 = a_6 = 0$ and $a_3 = 1$ to get $r = s^2, u^3 = 1, s = r^2, t^2 + t = r^3$. From this we see that $u \in \{1, \omega, \omega^2\}$ and $r \in \{0, 1, \omega, \omega^2\}$. Also $t^2 + t + r^3 = 0$ is always separable, so for r given t has two choices. In total we then get the 24 automorphisms. Now to show $\text{Aut}(E)$ is non-abelian, it suffices to show that two automorphisms do not commute.

Proposition 5.5. Let E be the elliptic curve $y^2 = x^3 - x$. Suppose there is a point $P \in E(\mathbb{Q})$ with non-zero y coordinate. Use this assumption to show that the equation $w^2 = uv(u - v)(u + v)$ has non-trivial solution over $\mathbb{Z}$ with u, v coprime and $u \not\equiv v \pmod{2}$.

Show that there are positive integers a, b, c, d satisfying $a^2 = u, b^2 = v, c^2 = u + v, d^2 = u - v$. Show further that $(\frac{c+d}{2}, \frac{c-d}{2}, a)$ is a primitive Pythagorean triple and that the area of the corresponding right-angled triangle is $(w')^2$, where $w' = b/2$. Deduce that this is impossible and hence $E(\mathbb{Q}) = \{\mathcal{O}_E, (0, 0), (1, 0), (-1, 0)\}$

Proof. Let $P = (x, y) \in E(\mathbb{Q})$ with $y \neq 0$. Then $x \neq 0$ and $x > 1$, or $-1 < x < 0$ in which case we replace P with $(-1/x, y/x^2)$. So we can assume $x > 1$. Let $x = u/v$ with $u > v > 0$ coprime integers. Then $y^2 v^4 = uv(u-v)(u+v) \in \mathbb{Z}$ and $yv^2 \in \mathbb{Q}$ implies that $yv^2 \in \mathbb{Z}$. Let $w = yv^2$ and we get the required form. And we get $u \not\equiv v \pmod{2}$ by taking $v_2(-)$ of RHS and LHS.

Since $u, v, u+v, u-v$ are pairwise coprime, $w^2 = uv(u-v)(u+v)$ implies that $a^2 = u, b^2 = v, c^2 = u+v, d^2 = u-v$ for some $a, b, c, d \in \mathbb{N}^+$. Then $(\frac{c+d}{2})^2 + (\frac{c-d}{2})^2 = u = a^2$, and $\frac{c+d}{2}$ and $\frac{c-d}{2}$ are coprime since c, d are coprime. So $(\frac{c+d}{2}, \frac{c-d}{2}, a)$ is a primitive Pythagorean triple with area $v/4 = b^2/4$.

Now by modulo 4, one of $\frac{c+d}{2}$ and $\frac{c-d}{2}$ is even, and further we can get new $u', v' \in \mathbb{N}^+$ such that $(\frac{c+d}{2}, \frac{c-d}{2}, a) = (u'^2 - v'^2, 2u'v', u'^2 + v'^2)$ and $u'v'(u'^2 - v'^2) = w^2$, so we get a new solution to the equation. Moreover $4w'^2 = v|w^2 \implies w' \leq w/2$. So this would get a contradiction if we put a minimality assumption on w initially.

In other word, we showed that E has no rational point with non-zero y coordinate. Letting $y = 0$, we see a rational point of E is precisely one of the four. $\square$

Remark 5.1. This is important because $y^2 = x^3 - x$ represents a class of elliptic curves whose rational points can be studied via 2-isogeny.

Corollary 5.3. Consider $C \subset \mathbb{P}^2$ cut out by $U^4 + V^4 = W^4$. Find all rational points on C

Letting $x = W^2/U^2$ and $y = V^2W/U^2$. We see that $y^2 = x^3 - x$. For $P = [U : W : V] \in C(\mathbb{Q})$, if $U = 0$, then $W^4 = V^4$ and hence $W = \pm V$. We get $[0 : 1 : 1]$ and $[0 : 1 : -1]$

If $U \neq 0$, by $x = W^2/U^2$ and $y = V^2W/U^2$ we get a rational point on E . Hence by proposition 5.5 $V^2W/U^2 = 0$ and so V or W is 0. The first case gives $[1 : 0 : 1]$ and $[-1 : 0 : 1]$, and when $W = 0$ there is no solution.

6 Elliptic Curves over $\mathbb{C}$

A **lattice** Λ in $\mathbb{C}$ is a $\mathbb{Z}$ -span of two $\mathbb{R}$ -linearly independent $\omega_1, \omega_2 \in \mathbb{C}$, i.e. $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$. The quotient $X_\Lambda = \mathbb{C}/\Lambda$ is a Riemann surface, which is indeed a genus-1 torus. We will see lattices are in one-to-one correspondence with elliptic curves over $\mathbb{C}$.

Definition 6.1. An elliptic function is $f : \mathbb{C} \rightarrow \mathbb{C}$ meromorphic such that $f(z + \lambda) = f(z)$ for any $z \in \mathbb{C}, \lambda \in \Lambda$

These are precisely the meromorphic functions on X_Λ which form a field $\mathbb{C}(X_\Lambda)$

Definition 6.2. The **Weierstrass $\wp$ -function** w.r.t Λ is $\wp(z) = \wp(z, \Lambda) = 1/z^2 + \sum_{\lambda \in \Lambda \setminus 0} (1/(z - \lambda)^2 - 1/\lambda^2)$ which is a elliptic function with derivative $\wp'(z) = -2 \sum_{\lambda \in \Lambda} 1/(z - \lambda)^3$

Proposition 6.1. 1. $\mathbb{C}(X_\Lambda)$ is spanned by $\wp(z)$ and $\wp'(z)$

2. For $k \geq 2$ define the **Eisenstein series** of weight $2k$ for Λ :

$$G_{2k}(\Lambda) = \sum_{\lambda \in \Lambda \setminus 0} \lambda^{-2k}$$

and set $g_2(\Lambda) = 60G_4(\Lambda)$, $g_3(\Lambda) = 140G_6(\Lambda)$. Then $E_\Lambda : y^2 = x^3 - \frac{1}{4}g_2(\Lambda)x - \frac{1}{4}g_3(\Lambda)$ defines an elliptic curves over $\mathbb{C}$ and $\phi_\Lambda : X_\Lambda \rightarrow E_\Lambda; z \mapsto (\wp(z), \frac{1}{2}\wp'(z)), 0 \mapsto \mathcal{O}_E$ is a group isomorphism

Theorem 6.1. (Uniformization) Given $E/\mathbb{C}$ an elliptic curve, there is a unique lattice Λ such that $E = E_\Lambda$. Moreover, $E(\mathbb{C}) \cong \mathbb{C}/\Lambda$

Corollary 6.1. $E[n] \cong \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^2$ and hence $\deg[n] = n^2$

We have not proved that $[n]$ is separable which is done by $[n]^*\omega = n\omega \neq 0$ for the invariant differential ω . Assuming this, note that each $z \in \mathbb{C}/\Lambda$ has n^2 pre-images under $[n]$. So this follows from theorem 4.3

7 Elliptic Curves over $\mathbb{R}$

Given $E/\mathbb{R} : y^2 = x^3 + ax + b$. Theorem 6.1 implies that there is a unique lattice $\Lambda \subset \mathbb{C}$ such that $E_\Lambda = E$. Let σ be the complex conjugation.

Lemma 7.1. $\sigma(\Lambda) = \Lambda$. Moreover, the diagram

$$\begin{array}{ccc} \mathbb{C}/\Lambda & \xrightarrow{\phi_\Lambda \sim} & E(\mathbb{C}) \\ \downarrow \sigma & & \downarrow \sigma \\ \mathbb{C}/\Lambda & \xrightarrow{\phi_\Lambda \sim} & E(\mathbb{C}) \end{array}$$

commutes.

Since σ is continuous and $g_2(\Lambda), g_3(\Lambda) \in \mathbb{R}$, $g_2(\sigma\Lambda) = \sigma g_2(\Lambda) = g_2(\Lambda)$, $g_3(\sigma\Lambda) = g_3(\Lambda)$. So $E_\Lambda = E_{\sigma\Lambda}$. By the uniqueness in theorem 6.1, we get $\Lambda = \sigma\Lambda$

Now σ continuous implies $\sigma\wp(z; \Lambda) = \wp(\sigma z; \sigma\Lambda) = \wp(\sigma z; \Lambda)$ and $\sigma\wp'(z; \Lambda) = \wp'(\sigma z; \Lambda)$. So the diagram commutes.

Remark 7.1. Lemma 7.1 implies that $\text{Gal}(\mathbb{C}/\mathbb{R}) = \{1, \sigma\}$ acts on $\Lambda \cong \mathbb{Z}^2$. And the following lemma helps to understand this

Lemma 7.2. Let $M \in GL_2(\mathbb{Z})$ of order 2. Then M is conjugate (in $GL_2(\mathbb{Z})$) to $\pm I$, $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ or $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

Theorem 7.1. $E(\mathbb{R}) \cong \mathbb{R}/\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ if $\Delta(E) > 0$ and $E(\mathbb{R}) \cong \mathbb{R}/\mathbb{Z}$ if $\Delta(E) < 0$

Proof. The diagram in lemma 7.1 shows that $E(\mathbb{R}) \cong \{z \in \mathbb{C}/\Lambda \mid z \equiv \sigma z \pmod{\Lambda}\}$. Lemma 7.2 shows that σ acts on $\Lambda \cong \mathbb{Z}^2$ by one of $\pm I$, $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Let ω_1, ω_2 be the $\mathbb{Z}$ -basis for Λ . Then $\omega_1, \omega_2 \in \mathbb{R}$ if σ acts as I and $\omega_1, \omega_2 \in i\mathbb{R}$ if σ acts by $-I$. Both contradicts that ω_1, ω_2 are $\mathbb{R}$ -linearly independent.

If σ acts as $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ then $\omega_1 \in \mathbb{R}$, $\omega_2 \in i\mathbb{R}$. Note that $\mathbb{C} = \mathbb{R} \oplus i\mathbb{R}$. Now for $z \in \mathbb{C}$, $z \equiv \sigma(z) \pmod{\Lambda}$ if and only if $z \in \mathbb{R}\omega_1 + \frac{1}{2}\mathbb{Z}\omega_2$. So $E(\mathbb{R}) \cong (\mathbb{R}\omega_1 + \frac{1}{2}\mathbb{Z}\omega_2)/(\mathbb{Z}\omega_1 + \mathbb{Z}\omega_2) \cong \mathbb{R}/\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$. This implies $E(\mathbb{R})[2] \cong (\mathbb{Z}/2\mathbb{Z})^2 = E[2]$. So 2-torsions have coordinates in $\mathbb{R}$. But the 2-torsions are of the form $(t, 0)$ with t a root of $f(x) = x^3 + ax + b$. So f has three real roots and $\Delta(E) > 0$

If σ acts as $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ then $\sigma(\omega_1) = \omega_2$ and hence for $z \in \mathbb{C}$, $z \equiv \sigma(z) \pmod{\Lambda}$ if and only if $z \pmod{\Lambda} \in \mathbb{R}(\omega_1 + \omega_2)/\Lambda$. So $E(\mathbb{R}) \cong \mathbb{R}/\mathbb{Z}$ and $E(\mathbb{R})[2] \cong \mathbb{Z}/2\mathbb{Z}$. So f only has one real root and $\Delta(E) < 0$ $\square$

8 Isogeny

Given K a perfect field, $E/K, E'/K$ two elliptic curves,

Definition 8.1. An *isogeny* over K $\phi : E \rightarrow E'$ is a non-constant morphism defined over K with $\phi(\mathcal{O}_E) = \mathcal{O}_{E'}$. In this case we say E and E' are isogenous.

Remark 8.1. If $\phi : E \rightarrow E'$ is a non-constant morphism, then $T_{-f(\mathcal{O}_E)} \circ \phi$ is an isogeny

It is clear that the composition $\varphi \circ \phi$ of two isogenies ϕ, φ is an isogeny. And by tower law $\deg(\varphi \circ \phi) = \deg(\phi)\deg(\varphi)$

Lemma 8.1. If $\phi : E \rightarrow E'$ is an isogeny, then ϕ is a homomorphism regarding $(E/K, \oplus)$

This follows from the commutative diagram

$$\begin{array}{ccc} E & \xrightarrow{\phi} & E' \\ \downarrow \alpha_1, \sim & & \downarrow \alpha_2, \sim \\ Cl^0(E) & \xrightarrow{\phi_*} & Cl^0(E') \end{array}$$

where $\alpha_1(P) = [(P) - (\mathcal{O}_E)]$, $\phi_*([\sum n_P(P)]) = [\sum n_P(\phi(P))]$. Previously we have seen α_1, α_2 are isomorphisms, while ϕ_* is a group homomorphism. This implies that ϕ is a group homomorphism as well.

Definition 8.2. We write $\text{Hom}(E, E')$ the group of isogenies $E \rightarrow E'$ over $\bar{K}$ with the zero map, and $\text{End}(E) = \text{Hom}(E, E)$ the endomorphism ring of E , with composition as multiplication

Remark 8.2. For abelian varieties in general, homomorphisms need not to be an isogeny. What makes this special is that elliptic curves are simple, i.e. contains no proper abelian sub-variety for dimension reason, so non-constant homomorphisms are forced to be surjective and hence isogenies.

Lemma 8.2. For $n \neq 0$, $[n]$ is an isogeny.

We assume $\text{char}(K) \neq 2$. It suffices to show that $[n] \neq 0$. For $n = 2$, we have seen that $\#|E[2]| = 4$ ($\mathcal{O}_E$ and roots of $f(x)$). So $[2] \neq 0$. For n odd, take $\mathcal{O}_E \neq T \in E[2]$, then $nT = T$ since $n \equiv 1 \pmod{2}$. So $[n] \neq 0$.

In general, we use $[mn] = [m] \circ [n]$

Corollary 8.1. $\text{Hom}(E, E')$ is $\mathbb{Z}$ -torsionfree as a $\mathbb{Z}$ -module

Take $f : E \rightarrow E'$ an isogeny. Then $nf = [n] \circ f$ is the composition of isogenies and hence an isogeny.

In the following content the lecturer uses many techniques with some very explicit calculation in order to show that $\deg([n]) = n^2$. We will skim most part and briefly recall how this is done in AEC using dual isogeny.

Let $\phi : E \rightarrow E'$ be an isogeny. There exists a unique isogeny $\widehat{\phi} : E' \rightarrow E$ satisfying $\widehat{\phi} \circ \phi = [m]$. Indeed, it is the composition $\alpha_1^{-1} \circ \phi^* \circ \alpha_2$ with the settings in lemma 8.1. This is called the **dual isogeny** of ϕ . The proof of the existence is divided into two cases, where ϕ is separable or a Frobenius (totally inseparable part) morphism. We will just use it freely.

Moreover, $\widehat{[m]} = [m]$. This comes from another non-trivial fact: $\widehat{\phi + \varphi} = \widehat{\phi} + \widehat{\varphi}$. Hence if $[m]$ is of degree d , then $[d] = \widehat{[m]} \circ [m] = [m] \circ [m] = [m^2]$. So $m^2 = d$ since $\text{End}(E)$ is torsionfree.

Back to the lecture notes

Definition 8.3. *A an abelian group. We say $q : A \rightarrow \mathbb{R}$ is a **quadratic form** if $q(nx) = n^2q(x)$ for all $x \in A$, $n \in \mathbb{N}$, and $\langle -, - \rangle_q : A \times A \rightarrow \mathbb{R}; (x, y) \mapsto q(x + y) - q(x) - q(y)$ is $\mathbb{Z}$ -bilinear.*

The rest of this section is built around the following theorem:

Theorem 8.1. *E, E' elliptic curves. Then $\deg : \text{Hom}(E, E') \rightarrow \mathbb{Z}$ is a quadratic form.*

We will omit its proof, but note that there is a criterion for quadratic form

Lemma 8.3. *$q : A \rightarrow \mathbb{R}$ is a quadratic form if and only it satisfies $q(x + y) + q(x - y) = 2q(x) + 2q(y)$ for all $x, y \in A$. This is called the parallelogram law.*

The proof of theorem 8.1 essentially shows that $\deg(\phi + \varphi) + \deg(\phi - \varphi) = 2\deg(\phi) + 2\deg(\varphi)$. We will recycle this in the section on elliptic curves over finite fields

By theorem 8.1, $\deg(n\phi) = n^2\deg(\phi)$, so we immediately get $\deg([n]) = n^2$

Our next aim is to show $E[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$. Recall that for $\text{char}(K) \neq 2$ $E : y^2 = f(x)$, the **invariant differential** $\omega = dx/2y$ is the unique (up to constant) regular differentials, i.e. $\text{div}(\omega) = 0$

Lemma 8.4. *Let $P \in E(K)$. Then $T_P^*\omega = \omega$*

Proof. T_P is an automorphism implies that $T_P^*\omega$ is regular and nonzero. So $T_P^*\omega = \lambda(P)\omega$ for some $\lambda \in K^\times$. Then $\lambda : E \rightarrow \mathbb{P}^1; P \mapsto \lambda(P)$ is a morphism. But $0, \infty$ are not in the images implies that λ is a constant. Since $\lambda(\mathcal{O}_E) = 1$, we see that $T_P^*\omega = \omega$ for all P . $\square$

Proposition 8.1. *Let $\phi, \varphi \in \text{Hom}(E, E')$ and ω the invariant differential of E' . Then $(\phi + \varphi)^*\omega = \phi^*\omega + \varphi^*\omega$*

This is another non-trivial fact that we will omit its proof.

Corollary 8.2. *$[n]^*\omega = n\omega \neq 0$ for $n \neq 0$. In particular, $[n]$ is separable.*

This is done by induction.

Theorem 8.2. *If $\text{char}(K) \nmid n$, then $E[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$*

Proof. by corollary 8.2, $[n]$ is separable, $\#[[n]^{-1}(Q)] = \deg([n]) = n^2$ for all but finitely many points. But in this case, by taking $R \in [n]^{-1}(Q)$, $T \mapsto T - R$ gives a bijection between $[n]^{-1}(Q)$ and $E[n]$. So $\#[[n]^{-1}(Q)] = n^2$ for all Q and hence $\#E[n] = n^2$.

Write $E[n] \cong \mathbb{Z}/d_1\mathbb{Z} \times \dots \times \mathbb{Z}/d_k\mathbb{Z}$ with $d_1|d_2|\dots|d_k$ and $\prod d_i = n$. Now for any $p|d_1$, $E[p]$ is the p -torsion part of $E[n]$ and hence $E[p] \cong (\mathbb{Z}/p\mathbb{Z})^k$. So $k = 2$. Now since $d_1|d_2|n$ and $d_1d_2 = n$, we must have $d_1 = d_2 = n$ $\square$

Remark 8.3. For $n = \text{char}(K) = p$, $[p]$ is inseparable. And we have $E[p^r] \cong \mathbb{Z}/p^r\mathbb{Z}$ if E is ordinary and $E[p^r] = 0$ if E is singular.

9 Elliptic Curves over Finite Fields

We start with another lemma on quadratic forms:

Lemma 9.1. *For A an abelian group and $q : A \rightarrow \mathbb{R}$ a positive definite quadratic form (i.e. $q(x) \geq 0$ with equality holds if and only if $x = 0$). Then*

$$|q(x + y) - q(x) - q(y)| \leq 2\sqrt{q(x)q(y)}$$

Proof. WLOG let $x \neq 0$. Then $0 \leq q(mx + ny) = \frac{1}{2} \langle mx + ny, mx + ny \rangle = q(mx) + mn \langle x, y \rangle + q(ny)$ for all $m, n \in \mathbb{Z}$. Hence $q(x)t^2 + \langle x, y \rangle t + q(y) \geq 0$ for any $t \in \mathbb{Q}$. So the discriminant $\langle x, y \rangle^2 - 4q(x)q(y) \leq 0 \implies |\langle x, y \rangle| \leq 2\sqrt{q(x)q(y)}$ as required. $\square$

Now for $K = \mathbb{F}_q$. For $E/\mathbb{F}_q$, we define the **frobenius endomorphism** $\phi_q : (x, y) \mapsto (x^q, y^q)$, which is easily seen well-defined since E is defined over $\mathbb{F}_q$

Lemma 9.2. $\phi = \phi_q$ is of degree q and inseparable. As a result, $1 - \phi$ is separable

Proof. $\deg(\phi) = q$ follows from a lemma we skimmed. Note that $\phi^* \omega = \phi^*(dx/(2y + a_1x + a_3)) = dx^q/(2y^q + a_1x^q + a_3) = q(\dots) = 0$. Hence ϕ is inseparable. Also $(1 - \phi)^* \omega = \omega - \phi^* \omega = \omega \neq 0$ by proposition 8.1. So $1 - \phi$ is separable $\square$

Theorem 9.1. (Hasse's theorem) We have $|\#|E(\mathbb{F}_q)| - 1 - q| < 2\sqrt{q}$

Proof. Let $\phi = \phi_q$. Notice that the frobenius map generates the absolute Galois group of $\mathbb{F}_q$. Hence $E(\mathbb{F}_q) = \{P \in E(\overline{\mathbb{F}_q}) | \phi(P) = P\}$. So $E(\mathbb{F}_q) = \ker(1 - \phi)$. But since $1 - \phi$ is separable and a group homomorphism, $\deg(1 - \phi) = \#|\ker(1 - \phi)|$. Now by lemma 9.1 and the fact that $\deg$ map is a positive definitive quadratic form, we have $|\deg(1 - \phi) - 1 - \deg(\phi)| \leq 2\sqrt{\deg(\phi)} \implies |\#|E(\mathbb{F}_q)| - 1 - q| \leq 2\sqrt{q}$ by lemma 9.2. $\square$

We define $\text{tr}(\phi) = \deg(\phi + 1) - \deg(\phi) - 1$. It is linear in ϕ since $\text{tr}\phi = \langle \phi, 1 \rangle$ where $\langle -, - \rangle$ is the bilinear form induced from the degree map which is a quadratic form

Lemma 9.3. For $\phi \in \text{End}(E)$, $\phi^2 - \text{tr}(\phi)\phi + \deg(\phi) = 0$

This can be done by considering $\phi_\ell \in \text{Aut}(T_\ell E)$ induced by ϕ . One can show that $\text{tr}(\phi_\ell) = \text{tr}(\phi)$, $\deg(\phi_\ell) = \deg(\phi)$, So the statement here is Cayley-Hamilton and isogeny theorem. In example sheet, it is done in a very unnatural and explicit way. Notice that this shows that the characteristic polynomial of ϕ_ℓ is in $\mathbb{Z}[x]$. This is true in general for abelian varieties and was proved by Wéil

Proposition 9.1. Let $a = 1 + q - \#|E(\mathbb{F}_q)| = \text{tr}\phi$ and α, β roots of $t^2 - at + q = t^2 - \text{tr}(\phi)t + \deg\phi$. Then $|\alpha| = |\beta| = q^{1/2}$ and $\#|E(\mathbb{F}_{q^n})| = 1 + q^n - \alpha^n - \beta^n$

Proof. Hasse's theorem implies the discriminant < 0 . So $\alpha = \bar{\beta}$ and hence their length are both $q^{1/2}$. Now let $\phi = \phi_q$. Then $a = \text{tr}(\phi)$ and hence $\phi^2 - a\phi + q = 0$ by lemma 9.3. Hence $\phi^{n+2} - a\phi^{n+1} + q\phi^n = 0 \implies \text{tr}(\phi^{n+2}) - a\text{tr}(\phi^{n+1}) + q\text{tr}(\phi^n) = 0$. By solving the second order differential equation, we have $\text{tr}(\phi^n) = \alpha^n + \beta^n$. Hence $\#|E(\mathbb{F}_{q^n})| = \deg(1 - \phi^n) = \deg(\phi^n) + 1 - \text{tr}(\phi^n) = 1 + q^n - \alpha^n - \beta^n$ $\square$

Recall that $-\log(1 - T) = \sum_{n=1}^{\infty} T^n/n$

Definition 9.1. The **Zeta function** of a variety $V/\mathbb{F}_q$ is $Z_V(T) = \exp(\sum_{n=1}^{\infty} \frac{\#|V(\mathbb{F}_{q^n})|T^n}{n})$

Proposition 9.2. For $E/\mathbb{F}_q$ elliptic curves and $a = \text{tr}\phi = 1 + q - \#|E(\mathbb{F}_q)|$, $Z_E(T) = \frac{1 - aT + qT^2}{(1 - T)(1 - qT)}$

Let α, β be the roots of $t^2 - at + q$. By proposition 9.1, $Z_E(T) = \exp(\sum_{n=1}^{\infty} \frac{(1 + q^n - \alpha^n - \beta^n)T^n}{n}) = \exp(-\log(1 - T) - \log(1 - qT) + \log(1 - \alpha T) + \log(1 - \beta T)) = \frac{(1 - \alpha T)(1 - \beta T)}{(1 - T)(1 - qT)} = \frac{1 - aT + qT^2}{(1 - T)(1 - qT)}$

For number fields, we can define the Dedekind-Zeta function, which is related to the Zeta function we have just computed by change of variable.

9.1 Exercise

Proposition 9.3. 1. Find all points over $\mathbb{F}_{13}$ on the elliptic curve $E : y^2 = x^3 + x + 5$ and show that they form a cyclic group. Find an example of an elliptic curve over $\mathbb{F}_{13}$ for which this group is not cyclic. Are there any examples where the group requires more than two generators?

2. Without listing its elements, find the order of $E(\mathbb{F}_{13^2})$ and determine whether this group is cyclic.

Proof. (1) We can compute $E(\mathbb{F}_{13}) = \{\mathcal{O}_E, (3, \pm 3), (7, \pm 2), (10, \pm 1), (12, \pm 4)\}$. So $E(\mathbb{F}_{13}) \cong C_9$ or $C_3 \times C_3$. To show its cyclic, we can compute its 3-torsion point. In example 3 we give the formula for double points. For $P = (x, y)$ 3-torsion, we need $x = \frac{f'(x) - 8xf(x)}{4f(x)} \implies 3x^4 + 6x^2 + 60x - 1 = 0$. Plugging the x -coordinate of points in $E(\mathbb{F}_{13})$, we find that $E(\mathbb{F}_{13})[3] = \{0, (-1, \pm 4)\}$. So $E(\mathbb{F}_{13})$ only has 3 3-torsion points and hence is cyclic.

As in previous exercises, we choose $E' : y^2 = x^3 - x$. In this case we still have $E'(\mathbb{F}_{13}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$

There is no elliptic curves here the group requires more than two generators. To see this, let $E(\mathbb{F}_{13}) \cong \mathbb{Z}/d_1\mathbb{Z} \times \dots \times \mathbb{Z}/d_n\mathbb{Z}$ with $d_i \geq 2$ and $d_1 \mid \dots \mid d_n$. Take any prime $p \mid d_1$, we see that $|E(\mathbb{F}_{13})[p]| = p^n$. Since $|E(\mathbb{F}_{13})[p]| \leq p^2 = \deg([p])$, we have $n \leq 2$

(2) Let $\phi : E \rightarrow E$ be the Frobenius endomorphism. We have $\text{tr}(\phi) = 1 + 13 - |E(\mathbb{F}_{13})| = 5$, and $\deg(1 + \phi) = 2 + 2\deg(\phi) - \deg(1 - \phi) = 2 + 26 - 9 = 19$. So $|E(\mathbb{F}_{169})| = \deg(1 - \phi^2) = \deg(1 - \phi)\deg(1 + \phi) = 171$. Since $E(\mathbb{F}_{13}) \subset E(\mathbb{F}_{169})$ is cyclic subgroup of order 9, we know that $E(\mathbb{F}_{169})$ is cyclic $\square$

Proposition 9.4. Let E_1 and E_2 be elliptic curves over $\mathbb{F}_q$, and let $\varphi : E_1 \rightarrow E_2$ be an isogeny defined over $\mathbb{F}_q$. Let ϕ_i be the q -power Frobenius on E_i . Show that $\varphi \circ \phi_1 = \phi_2 \circ \varphi$ and deduce that $|E_1(\mathbb{F}_q)| = |E_2(\mathbb{F}_q)|$.

We write the isogeny φ as $(x, y) \mapsto (\alpha(x, y), \beta(x, y))$. Then α, β are defined over $\mathbb{F}_q$. Hence $\varphi \circ \phi_1(x, y) = (\alpha(x^q, y^q), \beta(x^q, y^q)) = (\alpha(x, y)^q, \beta(x, y)^q) = \phi_2 \circ \varphi(x, y)$. We deduce that $\varphi \circ (1 - \phi_1) = (1 - \phi_2) \circ \varphi$. Taking degree and we get $|E_1(\mathbb{F}_q)| = \deg(1 - \phi_1) = \deg(1 - \phi_2) = |E_2(\mathbb{F}_q)|$

Proposition 9.5. Let $E/\mathbb{F}_p$ be an elliptic curve with p an odd prime. Show that there exists an elliptic curve $E'/\mathbb{F}_p$ with $|E(\mathbb{F}_p)| + |E'(\mathbb{F}_p)| = 2p + 2$. Show further that $E(\mathbb{F}_p) \times E'(\mathbb{F}_p)$ and $E(\mathbb{F}_{p^2})$ have the same order but need not to be isomorphic.

Proof. We should note that there is a naive way to present the number of points over $\mathbb{F}_p$ on $E : y^2 = f(x)$

$$|E(\mathbb{F}_p)| = p + 1 + \sum_{x=0}^{p-1} \left(\frac{f(x)}{p} \right)$$

where $\left(\frac{\cdot}{p} \right)$ is the Legendre symbol. Now we take the quadratic twist $E_d : y^2 = df(x)$ for some $0 \neq d \in \mathbb{F}_p^\times / \mathbb{F}_p^{\times 2}$. Then we compute

$$\begin{aligned} |E(\mathbb{F}_p)| + |E'(\mathbb{F}_p)| &= 2p + 2 + \sum_{x=0}^{p-1} \left(\left(\frac{f(x)}{p} \right) + \left(\frac{df(x)}{p} \right) \right) \\ &= 2p + 2 + \sum_{x=0}^{p-1} \left(\left(\frac{f(x)}{p} \right) + \left(\frac{d}{p} \right) \left(\frac{f(x)}{p} \right) \right) \\ &= 2p + 2 + \sum_{x=0}^{p-1} \left(\left(\frac{f(x)}{p} \right) - \left(\frac{f(x)}{p} \right) \right) \end{aligned}$$

$$= 2p + 2$$

Now note that $\deg(1 + \phi) + \deg(1 - \phi) = 2 + 2\deg(\phi) = 2 + 2p$ for ϕ the Frobenius. Since $\deg(1 - \phi) = |E(\mathbb{F}_p)|$, we get $\deg(1 + \phi) = |E_d(\mathbb{F}_p)|$. Then $E(\mathbb{F}_p) \times E'(\mathbb{F}_p)$ and $E(\mathbb{F}_{p^2})$ have the same order since $|E(\mathbb{F}_{p^2})| = \deg(1 - \phi^2) = \deg(1 - \phi)\deg(1 + \phi)$.

Again, if we take $E : y^2 = x^3 - x$, we see that $E(\mathbb{F}_p)[2] = E_d(\mathbb{F}_p)[2] \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (still those four points). Hence their product has a 2-torsion subgroup isomorphic to $(\mathbb{Z}/2\mathbb{Z})^4$. But by the second part of proposition 9.3, this is impossible for $E(\mathbb{F}_{p^2})$ $\square$

Proposition 9.6. *Let E be an elliptic curve over $\mathbb{F}_p$ (p a prime) with $|E(\mathbb{F}_p)| = p + 1 - a$ (i.e. $a = \text{tr}\phi$), and let $\phi : E \rightarrow E$ be the p -power Frobenius, Let $\varphi = [a] - \phi$.*

1. show that $\varphi \circ \phi = \phi \circ \varphi = [p]$
2. Show that if φ is separable then $E[p^r] \cong \mathbb{Z}/p^r\mathbb{Z}$ for all $r \geq 1$
3. Show that if $p \geq 5$ and $E[p] = 0$ then $|E(\mathbb{F}_p)| = p + 1$.

Proof. (1) Note that $\phi^2 - [\text{tr}\phi]\phi + \deg(\phi) = 0 \implies [p] = \phi([\text{tr}\phi] - \phi)$, and $\varphi = [\text{tr}\phi] - \phi$

(2) $[p^r] = \phi^r \circ \varphi^r$ since they commute. Since $\phi(x, y) = (x^p, y^p)$, $E[\phi] = 0$ and hence $E[p^r] = E[\phi^r \circ \varphi^r] = E[\varphi^r]$. Now $\deg(\varphi) = p$ since $\deg(\phi) = p$ and $\deg([p]) = p^2$. Also φ^r is separable since φ is separable (by considering $\varphi^*\omega \neq 0$). So $|E[p^r]| = |E[\varphi^r]| = \deg(\varphi^r) = p^r$ since φ^r is a homomorphism. To see $E[p^r] \cong \mathbb{Z}/p^r\mathbb{Z}$, note that $E[p^r][p] = E[p] = \mathbb{Z}/p\mathbb{Z}$

(3) If $E[p] = 0$, we know φ is inseparable by (2). Hence $\varphi^*\omega = ([a] - \phi)^*\omega = [a]^*\omega - \phi^*\omega = [a]^*\omega = 0$ since ϕ is inseparable. But $[a]^*\omega = a \cdot \omega = 0$ implies that $a = 0$ in $\mathbb{F}_p$, i.e. $p \mid a$. Now by Hensel's theorem $|a| \leq 2\sqrt{p}$. For $p \geq 5$, this implies that $a = 0$, and so $|E(\mathbb{F}_p)| = p + 1$. $\square$

10 Formal Group

The theory of formal groups is a main ingredient to study elliptic curves over local fields.

Definition 10.1. *Let R a commutative ring and $I \triangleleft R$. The **I-adic topology** on R is the topology with basis $\{r + I^n \mid r \in R, n \in \mathbb{N}^+\}$*

Definition 10.2. *A sequence (x_n) is **Cauchy** if for any $k \geq 1$, there is $N \in \mathbb{N}$ such that $x_m - x_n \in I^k$ for all $m, n \geq N$. R is **complete** w.r.t I-adic topology if $\bigcap_{n \geq 0} I^n = 0$ (Hausdorff) and every Cauchy sequence converges.*

Remark 10.1. R complete implies the natural homomorphism $R \rightarrow \varprojlim R/I^n$ is an isomorphism.

Lemma 10.1. *(Hensel's lemma) Given (R, I) complete, let $f \in R[x]$. Suppose $a \in R$ such that $f(a) \equiv 0 \pmod{I^s}$ for some $s \geq 1$ and $f'(a) \in R^\times$. Then there is a unique $b \in R$ such that $f(b) = 0$ and $b \equiv a \pmod{I^s}$*

Again, this is done by induction and Newton's method. See any notes on local fields.

Now consider an elliptic curve E/K given by the Weierstrass equation $y^2z + a_1xyz + a_3yz^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3$ with $\mathcal{O}_E = [0 : 1 : 0]$. Consider the affine chart $\{y \neq 0\}$ containing $\mathcal{O}_E$ as $(0, 0)$:

Set $t = -x/y$, $w = -z/y$, and we get $w = t^3 + a_1tw + a_2t^2w + a_3w^2 + a_4tw^2 + a_6w^3 =: f(t, w)$. And now t is a uniformizer at $\mathcal{O}_E$

Lemma 10.2. *There is a unique $w(t) \in \mathbb{Z}[a_1, \dots, a_6][[t]]$ such that $w(t) = f(t, w(t))$ and $w(t) \equiv 0 \pmod{t^3}$*

Proof. We apply Hensel's lemma to $R = \mathbb{Z}[a_1, \dots, a_6][[t]]$, $I = (t)$ and $F(x) = x - f(t, x)$. Check that $F(0) = -f(t, 0) = -t^3 \equiv 0 \pmod{I^3}$, $F'(0) = 1 - a_1 t - a_2 t^2$ is a unit. Hence by Hensel's lemma, there is a unique $w(t) \in \mathbb{Z}[a_1, \dots, a_6][[t]]$ such that $w(t) \equiv 0 \pmod{I^3}$ and $F(w(t)) = w(t) - f(t, w(t)) = 0$ as required. $\square$

Definition 10.3. Given (R, I) complete integral domain, $a_1, \dots, a_6 \in R$, $K = \text{Frac}(R)$, E/K an elliptic curve in Weierstrass form. Define $\widehat{E}(I) = \{(t_0, w_0) \in I \times I \mid w_0 = f(t_0, w_0)\}$ with $f(t, w)$ as above.

Remark 10.2. Let $\mathcal{O}_E = (0, 0) \neq P = (t_0, w_0) \in \widehat{E}(I)$. Then in the usual coordinate (i.e. in the affine chart $\{z \neq 0\}$), $(x, y) = (t/w, -1/w)$ satisfy the Weierstrass equation. In this way $\widehat{E}(I)$ is naturally a subset of $E(K)$.

Lemma 10.3. With the settings in definition 10.3, $\widehat{E}(I) = \{(t, w(t)) \mid t \in I\}$ and is a subgroup of $E(K)$.

Proof. For a fixed $t_0 \in I$ we can apply Hensel's lemma to $x - f(t_0, x)$ and see that there is a unique $w_0 \in I$ such that (t_0, w_0) satisfy the equation. But $w(t_0) \in I^3$ also meets the requirement, the uniqueness gives $w_0 = w(t_0)$.

Now since $\mathcal{O}_E = (0, 0) \in \widehat{E}(I)$, it suffices to show that for $P_1, P_2 \in \widehat{E}(I)$, $-P_1 - P_2 \in \widehat{E}(I)$ as well. Consider the line $L : w = \lambda t + v$ through P_1, P_2 . Then $\lambda = \frac{w(t_1) - w(t_2)}{t_1 - t_2}$. Since $w(t) \equiv 0 \pmod{t^3}$, $\lambda \in I$ and hence $v = w_1 - \lambda t_1 \in I$. Substituting $w = \lambda t + v$ in the equation, we get

$$f(t, \lambda t + v) - \lambda t - v = 0$$

Note LHS is a cubic in t . Let A be the coefficient of t^3 and B the coefficient of t^2 on the LHS. Modulo I on the LHS we get $f(t, 0) = t^3$. So $A \equiv 1 \pmod{I}$ i.e. a unit, $B \in I$. Then $t_3 = -B/A - t_1 - t_2 \in I$ and $w_3 = \lambda t_3 + v \in I$. So the third point of intersection $-P_1 - P_2 = (t_3, w_3) \in \widehat{E}(I)$ $\square$

Corollary 10.1. 1. There exists $i(t) \in \mathbb{Z}[a_1, \dots, a_6][[t]]$ with $i(t) \in (t)$ such that $(i(t), w(i(t))) = \ominus(t, w(t))$. In fact, i can be given by explicit formula $i(t) = -t - a_1 t^2 - a_2 t^3 - (a_1^3 + a_3) t^4 \dots$

2. There exists $F(t_1, t_2) \in \mathbb{Z}[a_1, \dots, a_6][[t_1, t_2]]$ with $F(t_1, t_2) \in (t_1, t_2)$ such that $(t_1, w(t_1)) \oplus (t_2, w(t_2)) = (F(t_1, t_2), w(F(t_1, t_2)))$. In fact $F(x, y) = x + y - a_1 xy - a_2(x^2 + xy + y^2) + \dots$

This is clear by lemma 10.3 and taking $R = \mathbb{Z}[a_1, \dots, a_6][[t]]$, $I = (t)$ and $R = \mathbb{Z}[a_1, \dots, a_6][[t_1, t_2]]$, $I = (t_1, t_2)$ separately.

Since $\widehat{E}(I)$ is an abelian group (as it is a subgroup of $E(K)$), we deduce that

1. $F(X, 0) = X$, $F(0, Y) = 0$
2. $F(X, Y) = F(Y, X)$
3. $F(F(X, Y), Z) = F(X, F(Y, Z))$ by associativity
4. $F(x, i(X)) = 0$

This leads to the following definition

Definition 10.4. Given R a ring, a (one parameter commutative) formal group law over R is a power series $F(X, Y) \in R[[X, Y]]$ satisfies the properties above. Moreover, if (R, I) complete, for $x, y \in I$, we write $x +_F y = F(x, y) \in I$, which gives a new addition $+_F$ on I . Denote $(I, +_F)$ by $F(I)$

Remark 10.3. One can show that for a power series F satisfying 1, 2, 3 above, there is a unique $i(x) = -x + \dots \in R[[X]]$ such that $F(X, i(X)) = 0$.

Example 6. 1. $F(X, Y) = X + Y$. This is the formal additive group $\widehat{\mathbb{G}}_a$. $F(I) = (I, +)$

2. $F(X, Y) = X + Y + XY$. This is the formal multiplicative group $\widehat{\mathbb{G}}_m$. $F(I) = (1 + I, \times)$

3. $F(X, Y)$ in corollary 10.1 (2) constructed from an elliptic curve with coefficients in R . We denote it by $\widehat{E}$, the formal group of E . The group law is consistent with corollary 10.1

Definition 10.5. Given F, G two formal group laws over R , a homomorphism $\theta : F \rightarrow G$ is a power series $\theta \in R[[T]]$ satisfying $\theta(0) = 0$ and $\theta(F(X, Y)) = G(\theta(X), \theta(Y))$. We say that $F \cong G$ if there are homomorphisms $\theta : F \rightarrow G$ and $\phi : G \rightarrow F$ satisfying $\theta(\phi(T)) = \phi(\theta(T)) = T$

Remark 10.4. For (R, I) complete, θ as above induces an homomorphism $F(I) = (I, +_F) \rightarrow G(I) = (I, +_G)$ since $\theta(x +_F y) = \theta(x) +_G \theta(y)$ for $x, y \in I$ by definition.

We define $[n] : F \rightarrow F$ the multiplication by n homomorphism, by $[1](T) = T$, $[n](T) = F(T, [n-1]T)$ and $[-n](T) = i \circ [n](T)$

Lemma 10.4. Given R a ring, $a \in R^\times$ and $\theta(T) = aT + (\deg \geq 2 \text{ part}) \in R[[T]]$. Then there exists a unique $\phi(T) = a^{-1}T + \dots \in R[[T]]$ such that $\phi(\theta(T)) = \theta(\phi(T)) = T$

This is by constructing a sequence (ϕ_n) converging to ϕ such that $\theta(\phi_n(T)) \equiv T \pmod{T^{n+1}}$ and hence $\theta(\phi(T)) = T$. Similarly we get $\varphi \in R[[t]]$ with $\varphi(\theta(T)) = T$. Then $\varphi(T) = \varphi(\theta(\phi(T))) = \phi(T)$

Lemma 10.5. Given $\theta : F \rightarrow G$ a homomorphism, if there is $\phi \in R[[t]]$ with $\phi(0) = 0$ such that $\phi(\theta(T)) = \theta(\phi(T)) = T$. Then ϕ is a homomorphism $G \rightarrow F$ and hence an isomorphism.

In particular, given θ a homomorphism, by lemma 10.4 & 10.5 θ is an isomorphism if and only if $\theta'(0) \in R^\times$

Proposition 10.1. Let F/R a formal group, $n \in \mathbb{Z}$, $n \in R^\times$. Then $[n] : F \rightarrow F$ is a isomorphism. In particular, for (R, I) complete, $F(I) \xrightarrow{n} F(I)$ is an isomorphism, which shows that $F(I)$ is torsionfree

Since $F(X, Y) = X + Y + (\deg \geq 2 \text{ part})$, we can prove by induction that $[n]T = nT + (\deg \geq 2 \text{ part})$. So this follows the observation above.

Now we give a classification of formal groups over characteristic 0

Theorem 10.1. If $\text{char}(R) = 0$, then any formal group F/R is isomorphic to $\widehat{\mathbb{G}}_a$ over $R \otimes_{\mathbb{Z}} \mathbb{Q}$. More precisely

1. There is a unique power series $\log(T) = T + a_2 T^2/2 + \dots, a_i \in R$ such that $\log(F(X, Y)) = \log(X) + \log(Y)$

2. There exists a unique power series $\exp(T) \in R[[T]]$ such that $\log(\exp(T)) = \exp(\log(T)) = T$

Then by lemma 10.5, $\log$ is an isomorphism between F and $\widehat{\mathbb{G}}_a$

Proof. Let $F_1(X, Y) = \partial F / \partial X(X, Y)$. For uniqueness, suppose $\log(T)$ is taken. Let $P(T) = d(\log(T))/dT = 1 + a_2 T + \dots$. We take $\partial / \partial X$ on both sides of $\log(F(X, Y)) = \log(X) + \log(Y)$ and get $F_1(X, Y)P(F(X, Y)) = P(X)$. Now take $X = 0$, and we get $F_1(0, Y)P(Y) = P(0) = 1$. So $P(Y) = F_1(0, Y)^{-1}$. Thus $d(\log(T))/dT$ is unique and hence $\log(T)$ is unique.

For existence, we define $P(T) = F_1(0, T)^{-1} = 1 + a_2 T + \dots$ and $\log(T) = T + a_2 T^2/2 + \dots$. Recall that $F(F(X, Y), Z) = F(X, F(Y, Z))$. Take $\partial / \partial X$ and we get $F_1(X, Y)F_1(F(X, Y), Z) = F_1(X, F(Y, Z))$. Again, set $X = 0$, we get $P(Y)^{-1}F_1(Y, Z) = P(F(Y, Z))^{-1}$, i.e. $F_1(Y, Z)P(F(Y, Z)) = P(Y)$. now integrate w.r.t y , we have $\log(F(Y, Z)) = \log(Y) + h(Z)$ for some $h(T) \in R[[T]]$. But Y and Z are symmetric on LHS implies that $\log(Z) = h(Z)$. This gives (1).

For (2), this follows from lemma 10.4. □

11 Elliptic Curves over Local Fields

In this section K is complete w.r.t a complete valuation $v : K^\times \rightarrow \mathbb{Z}$. Let $\mathcal{O}_K$ the ring of integers and $\mathcal{O}_K^\times$ the unit group. Let π be the uniformizer and k the residue field. Assume $\text{char}(K) = 0$, $\text{char}(k) = p$ and k perfect.

Definition 11.1. Let E/K an elliptic curve. A Weierstrass equation for E is **integral** if $a_1, \dots, a_6 \in \mathcal{O}_K$. An integral Weierstrass equation is **minimal** if $v(\Delta)$ is minimal amongst all integral Weierstrass equations, where $\Delta = \Delta(a_1, \dots, a_6)$ is the discriminant.

Remark 11.1. For the change of variable $x' = u^2x, y' = u^3y, a'_i = u^i a_i$ and hence the integral Weierstrass equation exists. Moreover, $a_1, \dots, a_6 \in \mathcal{O}_K \implies \Delta \in \mathcal{O}_K$ and hence $v(\Delta) \geq 0$. So *minimal equations exist*.

For a general change of variable $x' = u^2x + s, y' = u^3y + ru^2x + t$, we have $\Delta' = u^{12}\Delta$. In particular, for an integral Weierstrass equation, if $0 \leq v(\Delta) \leq 12$ then it is minimal. *The converse is not true.*

If $\text{char}(k) \neq 2, 3$, then there is a minimal Weierstrass equation of the form $y^2 = x^3 + ax + b$ for $a, b \in \mathcal{O}_K$.

Definition 11.2. The **reduction** $\bar{E}/k$ of E/K is the (projective) curve over k defined by reducing the coefficients of a minimal Weierstrass equation of E modulo π .

Lemma 11.1. The reduction $\bar{E}/k$ is well-defined. That is, given two minimal Weierstrass equation for E , their reductions modulo π are isomorphic over K .

Proof. Suppose these two equations are related by a change of variable $x' = u^2x + s, y' = u^3y + ru^2x + t$. Then $v(\Delta') = v(\Delta) + 12v(u) = v(\Delta) \implies u \in \mathcal{O}_K^\times$. Now the explicit transformation formula for a_i, a'_i and $\mathcal{O}_K$ integrally closed implies that $r, s, t \in \mathcal{O}_K$. Hence the reductions are isomorphic via the change of variable corresponding to $\bar{u} \in k^\times, \bar{r}, \bar{s}, \bar{t}$ $\square$

Definition 11.3. We say E has **good reduction** if $\bar{E}$ is non-singular, i.e. $\bar{\Delta}_{\min} \neq 0$, or $v(\Delta_{\min}) = 0$ where $\Delta_{\min}$ is the discriminant of a minimal Weierstrass equation.

Otherwise, $\bar{E}$ is singular and we have the results from §3 to apply.

Definition 11.4. We define a reduction map $\mathbb{P}^2(K) \rightarrow \mathbb{P}^2(k)$ as follows: represent P as $[x : y : z]$ where $0 = \min\{v(x), v(y), v(z)\}$. Set $\bar{P} = [\bar{x} : \bar{y} : \bar{z}]$. This induces $E(K) \rightarrow \bar{E}(k)$ given by $P \mapsto \bar{P}$.

We define $E_0(K) = \{P \in E(K) | \bar{P} \in \bar{E}_{ns}(k)\}$

Proposition 11.1. $E_0(K)$ is a subgroup of $E(K)$ and the reduction map gives a surjective homomorphism $E_0(K) \rightarrow \bar{E}_{ns}(k)$

Proof. We consider any line $L : ax + by + cz = 0 \subset \mathbb{P}_K^2$. WLOG assume $0 = \min\{v(a), v(b), v(c)\}$. Then modulo π it give a line $\bar{L} : \bar{a}x + \bar{b}y + \bar{c}z = 0 \subset \mathbb{P}_k^2$.

If $P_1, P_2, P_3 \in E(K)$ and $P_1 + P_2 + P_3 = \mathcal{O}_E$, they are colinear and lie on a line, say L . Then $\bar{P}_1, \bar{P}_2, \bar{P}_3 \in \bar{E}(k)$ lie on $\bar{L}$. Moreover, if $\bar{P}_1, \bar{P}_2$ are non-singular, then $\bar{P}_3$ is non-singular by Bezout's theorem. So $P_3 \in E_0(K)$, and hence $E_0(K)$ is a subgroup.

To show surjectivity, write the min. equation for E as $g(x, y) = 0$. Let $\bar{P} \in \bar{E}_{ns}(k) \setminus \mathcal{O}_{\bar{E}}$. Take x_0, y_0 any lifts of its coordinates. Then either $\partial g / \partial x(x_0, y_0) \not\equiv 0 \pmod{\pi}$ or $\partial g / \partial y(x_0, y_0) \not\equiv 0 \pmod{\pi}$. WLOG assume it is the first case. Let $G(t) = g(t, y_0)$. Then $G'(x_0) \in \mathcal{O}_K^\times$ and $G(x_0) \equiv 0 \pmod{\pi}$. By Hensel's lemma, there is $x \in \mathcal{O}_K$ such that $G(x) = g(x, y_0) = 0$ and $x \equiv x_0 \pmod{\pi}$, i.e. $(x, y_0) \in E(K)$ is a lift of $\bar{P}$ $\square$

Definition 11.5. Define $E_1(K) = \{P \in E(K) | \overline{P} = \mathcal{O}_E\} = \ker(E_0(K) \xrightarrow{P \mapsto \overline{P}} \overline{E}_{ns}(k))$

By proposition 11.1, $E_0(K)/E_1(K) \cong \overline{E}_{ns}(k)$. Inparticular, if k is a finite field, $E_1(K)$ has finite index in $E_0(K)$.

Lemma 11.2. If k is a finite field, then $E_0(K)$ has fintie index in $E(K)$

Sketch of proof. Since k is finite, by completeness $\mathcal{O}_K \cong \varprojlim \mathcal{O}_K/m^n$ is a pro-finite group and hence compact. Then $\mathbb{P}_K^n$ can be covered by finitely many compact sets $\{[a_1 : \dots : 1 : \dots : a_{n+1}] | a_i \in \mathcal{O}_K\}$ and hence compact. Then $E(K)$ is the cut-out of an equation and hence closed in $\mathbb{P}_K^2$ and so $E(K)$ is compact.

If $\overline{E}$ is singular with singular point $(\overline{x}_0, \overline{y}_0)$. Then $E(K) \setminus E_0(K) = \text{red}^{-1}(\overline{x}_0, \overline{y}_0)$ is closed. Hence $E_0(K)$ is open and its cosets form a disjoint open covering of $E(K)$ and hence there are only finitely many of them. So the index is finite. $\square$

Note that $P = (x, y) = [x : y : 1] \in E(K)$ reduces to $(\overline{x}, \overline{y})$ if $x, y \in \mathcal{O}_K$ and $\mathcal{O}_{\overline{E}}$ otherwise. This follows from the following lemma

Lemma 11.3. Let E/K with an integral Weierstrass equation $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$ and $\mathcal{O}_E \neq P = (x, y) \in E(K)$. Then either $x, y \in \mathcal{O}_K$ or $v(x) = -2s, v(y) = -3s$ for some $s \geq 1$

Proof. This is easily seen from the two sides of the equation. Suppose $v(x) \geq 0$ and $v(y) < 0$. Then y^2 dominates on the LHS and so $v(LHS) < 0$, while every term on the RHS is in $\mathcal{O}_K$ and so $v(RHS) \geq 0$, contradiction. So if $x \in \mathcal{O}_E, y \in \mathcal{O}_E$ as well.

Now suppose $v(x) < 0$. Then $v(RHS) = v(x^3) = 3v(x)$. Then $v(x) > v(y)$ since otherwise $v(LHS) \geq \min\{y^2, xy, y\} \geq 2v(x) > v(LHS)$. Then we see y^2 dominates LHS and so $v(LHS) = 2v(y)$. So $2v(y) = 3v(x)$, and the result follows. $\square$

Now note that K is complete implies that $\mathcal{O}_K$ is complete w.r.t $\pi^r \mathcal{O}_K$ for all $r \geq 1$. Now we fix a minimal Weierstrass equation for E/K with the associated formal group $\widehat{E}/\mathcal{O}_K$. Then we take $I = \pi^r \mathcal{O}_K$ in lemma 10.3 and get that $\widehat{E}(\pi^r \mathcal{O}_K) = \{(x, y) \in E(K) | v(-x/y) \geq r, v(-1/y) \geq r\} = \{(x, y) \in E(K) | v(x) \leq -2r, v(y) \leq -3r\}$ by variable transformation and lemma 11.3. Note that $E_1(K) = \widehat{E}(\pi \mathcal{O}_K)$. We define $E_r(K) = \widehat{E}(\pi^r \mathcal{O}_K)$

Proposition 11.2. If $\text{char}(k) \nmid n$, then multiplication by n is an isomorphism on $E_r(K)$ for all $r \geq 1$

Proof. $\text{char}(k) \nmid n$ implies that $n \in \mathcal{O}_K^\times$. So this follows from proposition 10.1 $\square$

Theorem 11.1. Let $F/\mathcal{O}_K$ a formal group. Let $e = v(p) > 0$. If $r > e/(p-1)$, then $\log : F(\pi^r \mathcal{O}_K) = (\pi^r \mathcal{O}_K, +_F) \rightarrow \widehat{\mathbb{G}}_a(\pi^r \mathcal{O}_K)$ is an isomorphism with inverse $\exp : \widehat{\mathbb{G}}_a(\pi^r \mathcal{O}_K) \rightarrow F(\pi^r \mathcal{O}_K)$. Note that $\widehat{\mathbb{G}}_a(\pi^r \mathcal{O}_K)$ is just $(\pi^r \mathcal{O}_K, +) \cong (\mathcal{O}_K, +)$

Proof. By theorem 10.1, it suffices to show that $\log(x)$ and $\exp(x)$ converge to an element in $\pi^r \mathcal{O}_K$ for $x \in \pi^r \mathcal{O}_K$. Recall that $\log(T) = T + a_2 T^2/2 + \dots, a_i \in R$. We claim that $v_p(n!) \leq (n-1)/(p-1)$. Indeed, $v_p(n!) = \sum [n/p^i] < \sum n/p^i = n/(p-1) \implies v_p(n!)(p-1) \leq n-1 < n$. So $v_p(n!)(p-1) \leq n-1$. Now claim gives $v(a_n x^n/n!) \geq nr - e v_p(n!) \geq nr - e(n-1)/(p-1) = (n-1)(r - e/(p-1)) + r$. By assumption, this goes to ∞ when $n \rightarrow \infty$. So $\log(x)$ converges. Moreover, each term has valuation at least r , so $\log(x) \in \pi^r \mathcal{O}_K$. Same for $\exp$ $\square$

Lemma 11.4. $F/\mathcal{O}_K$ formal group. Then for all $r \geq 1$, $F(\pi^r \mathcal{O}_K)/F(\pi^{r+1} \mathcal{O}_K) \cong (k, +)$

Proof. This is almost the same as showing $(\pi^r \mathcal{O}_K / \pi^{r+1} \mathcal{O}_K, +) \cong (k, +)$. Recall that $F(X, Y) = X + Y + (\deg \geq 2 \text{ part})$. So $F(\pi^r x, \pi^r y) \equiv \pi^r(x + y) \pmod{\pi^{r+1}}$. Thus the map $F(\pi^r \mathcal{O}_K) \rightarrow k; \pi^r x \mapsto \bar{x}$ is a surjective homomorphism with kernel $F(\pi^{r+1} \mathcal{O}_K)$ $\square$

Corollary 11.1. *$F/\mathcal{O}_K$ formal group and k a finite field. Then $F(\pi \mathcal{O}_K)$ has a finite index subgroup isomorphic to $(\mathcal{O}_K, +)$.*

By theorem 11.1, $F(\pi^r \mathcal{O}_K) \cong (\mathcal{O}_K, +)$ for $r > e/(p-1)$. The filtration from $F(\pi \mathcal{O}_K)$ to $F(\pi^r \mathcal{O}_K)$ has each quotient isomorphic to $(k, +)$ by lemma 11.4. So $F(\pi^r \mathcal{O}_K)$ has finite index.

Now we get back to elliptic curves.

Theorem 11.2. *E/K an elliptic curve over a local field K with $[K : \mathbb{Q}_p] < \infty$. Then $E(K)$ contains a subgroup of finite index isomorphic to $(\mathcal{O}_K, +)$. Moreover, $E(K)_{\text{tor}}$ is finite*

By consider the filtration of $E_r(K) = \widehat{E}(\pi^r \mathcal{O}_K)$, $E_1(K)$ contains a subgroup of finite index isomorphic to $(\mathcal{O}_K, +)$ by corollary 11.1 and $E_1(K)$ is of finite index by lemma 11.2 and $E_0(K)/E_1(K) \cong \overline{E}_{ns}(k)$. Now for $r > e/(p-1)$, $E_r(K) \cong (\mathcal{O}_K, +)$ is torsion free. Hence $E(K)_{\text{tor}} \hookrightarrow E(K)/E_r(K)$ which is finite since we essentially just showed that $E_r(K) \subset E(K)$ is of finite index.

Next we consider the extension of p -adic fields. Recall that for K with $[K : \mathbb{Q}_p] < \infty$, it has a unique unramified extension K_n of degree n for each $n \in \mathbb{N}^+$ (by adjoining the root of unity ζ_{q^n-1} if the residue field of K is $\mathbb{F}_q$). The **maximal unramified extension** of K

$$K^{nr} = \cup_{n \in \mathbb{N}^+} K_n$$

We use $[n]^{-1}Q$ to denote the preimages of Q under $[n]$ and $K(\{P_1, \dots, P_n\})$ the extension of K by adjoining the coordinates of the $\overline{K}$ -points P_i

Theorem 11.3. *Suppose E/K has good reduction and $p \nmid n$. Then $K([n]^{-1}P)/K$ is unramified*

Proof. Notice that good reduction gives a s.e.s $0 \rightarrow E_1(K_m) \rightarrow E(K_m) \rightarrow \overline{E}(k_m) \rightarrow 0$ for each m . Take $\cup_{m \geq 1}$ and we get

$$0 \rightarrow E_1(K^{nr}) \rightarrow E(K^{nr}) \rightarrow \overline{E}(\overline{k}) \rightarrow 0$$

since k is perfect.

By self-multiplication by n , we get a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & E_1(K^{nr}) & \longrightarrow & E(K^{nr}) & \longrightarrow & \overline{E}(\overline{k}) \longrightarrow 0 \\ & & \downarrow [n] & & \downarrow [n] & & \downarrow [n] \\ 0 & \longrightarrow & E_1(K^{nr}) & \longrightarrow & E(K^{nr}) & \longrightarrow & \overline{E}(\overline{k}) \longrightarrow 0 \end{array}$$

from this s.e.s. By proposition 10.1, the first vertical arrow is an isomorphism. Moreover, since $\overline{k}$ is algebraically closed, $[n]$ on $\overline{E}(\overline{k})$ is surjective with kernel isomorphic to $(\mathbb{Z}/n\mathbb{Z})^2$. By snake's lemma, we have $E(K^{nr})[n] \cong \overline{E}(\overline{k})[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$ and $E(K^{nr})/nE(K^{nr}) = 0$. This shows that $E[n] \subset E(K^{nr})$, i.e. the coordinates of n -torsions are in K^{nr} and $[n]$ on $E(K^{nr})$ is surjective. In particular, for $P \in E(K)$ there is $Q \in E(K^{nr})$ such that $nQ = P$. So for $P \in E(K)$ $[n]^{-1}P = \{Q + T | T \in E[n]\} \subset E(K^{nr})$ and so $K([n]^{-1}P) \subset K^{nr}$ $\square$

Remark 11.2. This theorem says that if E/K has good reduction then $E[n]$ for $p \nmid n$ have coords in K^{ur} , i.e. I acts trivially on them. Conversely, if I acts trivially on some Tate module $T_\ell(E)$ with $\ell \neq p$, then E has good reduction. This is the celebrated **Neron's criterion**.

12 Elliptic Curves over Number Fields

Given $[K : \mathbb{Q}] < \infty$ and E/K , we say E has good reduction over a place $\mathfrak{p}$ if $E/K_{\mathfrak{p}}$ has good reduction.

Lemma 12.1. *E/K has only finite places of bad (i.e. not good) reduction*

Proof. With the change of variables $x' = u^2x$ and $y' = u^3y$, we may assume $a_i \in \mathcal{O}_K$ for the Weierstrass equation of E . Then the discriminant $\Delta \in \mathcal{O}_K$ and $(\Delta) = \prod \mathfrak{p}^{c_i}$ for $c_i > 0$ and places $\mathfrak{p}_i$. For $\mathfrak{p}$ not in those finitely many $\mathfrak{p}_i$, $v_{\mathfrak{p}}(\Delta) = 0$, and hence $\overline{\Delta} \neq 0$ in $k_{\mathfrak{p}}$. So E has good reduction over $\mathfrak{p}$ $\square$

Lemma 12.2. *$E(K)_{\text{tor}}$ is finite. Moreover, for $\mathfrak{p}$ a place of good reduction with $\mathfrak{p} \nmid n$, the reduction map $E(K)[n] \hookrightarrow \overline{E}(k_{\mathfrak{p}})[n]$ is an injective homomorphism*

Proof. Note that $E(K)_{\text{tor}} \subset E(K_{\mathfrak{p}})_{\text{tor}}$ which is finite by theorem 11.2. Now since E has good reduction over $\mathfrak{p}$, proposition 11.1 says that the reduction map $E(K_{\mathfrak{p}}) \rightarrow \overline{E}(k_{\mathfrak{p}})$ is a surjective homomorphism with kernel $E_1(K_{\mathfrak{p}})$. By taking n -torsions we get a homomorphism $E(K_{\mathfrak{p}})[n] \rightarrow \overline{E}(k_{\mathfrak{p}})[n]$. Note that $[n]$ is an isomorphism on $E_1(K_{\mathfrak{p}})$ by proposition 10.1, so $E(K_{\mathfrak{p}})[n] \cap E_1(K_{\mathfrak{p}}) = 0$ and the homomorphism is injective. By composition we get $E(K)[n] \subset E(K_{\mathfrak{p}})[n] \hookrightarrow \overline{E}(k_{\mathfrak{p}})[n]$ $\square$

This lemma helps us to determine $\#|E(\mathbb{Q})_{\text{tor}}|$. For an abelian finite group and prime p , we know it is the product of p -primary part and p -prime part. By the lemma, we know the p -prime part of $E(\mathbb{Q})_{\text{tor}}$ is a subgroup of the p -prime part of $\overline{E}(\mathbb{F}_p)$, which is a subgroup of the finite group $\overline{E}(\mathbb{F}_p)$. On the other hand, the p -primary part of $E(\mathbb{Q})$ should have p -power order. This shows that $\#|E(\mathbb{Q})_{\text{tor}}|$ divides $\overline{E}(\mathbb{F}_p) \times$ some power of p .

Example 7. Consider $E/\mathbb{Q} : y^2 + y = x^3 - x^2$. The discriminant $\Delta = -11$, so E has good reduction at $p \neq 11$. We can compute that $\#|\overline{E}(\mathbb{F}_2)| = 5$ and $\#|\overline{E}(\mathbb{F}_3)| = 5$. So $\#|E(\mathbb{Q})_{\text{tor}}|$ divides 5. Note that $(0, 0) \in E(\mathbb{Q})$ has order 5. So $\#|E(\mathbb{Q})_{\text{tor}}| = 5$

Lemma 12.3. *Let $E/\mathbb{Q}$ given by a Weierstrass equation with $a_i \in \mathbb{Z}$. Suppose $\mathcal{O}_E \neq T = (x_0, y_0)$ is a torsion point. Then $4x_0, 8y_0 \in \mathbb{Z}$. If $2|a_1$ (e.g. the short Weierstrass form since $a_1 = 0$) or $2T \neq 0$, the $x_0, y_0 \in \mathbb{Z}$*

Note that the Weierstrass equation defines a formal group $\widehat{E}/\mathbb{Z}$. For $r \geq 1$, $\widehat{E}(p^r\mathbb{Z}_p) = \{(x, y) \in E(\mathbb{Q}) | v_p(x) \leq -2r, v_p(y) \leq -3r\}$. Theorem 11.1 says that $\widehat{E}(p^r\mathbb{Z}_p) \cong (\mathbb{Z}_p, +)$ which is torsionfree for $r > 1/(p-1)$. So $T \notin \widehat{E}(p^r\mathbb{Z}_p)$ for $r > 1/(p-1)$. Hence for p odd, $v_p(x_0), v_p(y_0) \geq 0$, and $v_2(x_0) \geq -2, v_2(y_0) \geq -3$.

Now suppose $T \in \widehat{E}(2\mathbb{Z}_2)$ so $v_2(x_0) = -2, v_2(y_0) = -3$. Then $\widehat{E}(2\mathbb{Z}_2)/\widehat{E}(4\mathbb{Z}_2) \cong (\mathbb{F}_2, +)$ by lemma 11.4. Then $2T \in \widehat{E}(4\mathbb{Z}_2)$ which is torsion free. So $2T = 0$ and $(x_0, y_0) = T = -T = (x_0, -a_1x_0 - a_3 - y_0)$. So $2y_0 + a_1x_0 + a_3 = 0$. By looking at the valuation at 2 of each term, we must have $v_2(a_1) = 0$ (otherwise, $2y_0$ dominates), i.e. $2 \nmid a_1$

Theorem 12.1. (Nagell-Lutz) *Suppose $E/\mathbb{Q} : y^2 = x^3 + ax + b = f(x)$ with $a, b \in \mathbb{Z}$ and $0 \neq T = (x_0, y_0)$ torsion point. Then $x_0, y_0 \in \mathbb{Z}$ and either $y_0 = 0$ or $y_0^2 \mid 4a^3 + 27b^2$*

Proof. Lemma 12.3 gives $x_0, y_0 \in \mathbb{Z}$. Suppose $y_0 \neq 0$, then $2T \neq 0$ and is torsion. Let $2T = (x_1, y_1)$. Then $x_1, y_1 \in \mathbb{Z}$. Now explicit formula gives $x_1 = (f'(x_0)/2y_0)^2 - 2x_0$. Euclidean's algorithm gives $(3x^2 + 4a)f'(x)^2 - 27(x^3 + ax - b)f(x) = 4a^3 + 27b^2$. Now $x_0, x_1 \in \mathbb{Z}$ implies $y_0^2 = f(x_0)|f'(x_0)$. By the second equations we get $y_0^2 \mid 4a^3 + 27b^2$ $\square$

Remark 12.1. This shows that the torsion points on $E/\mathbb{Q}$ are highly constrained.

12.1 Exercise

Proposition 12.1. Consider the elliptic curve $E/\mathbb{Q} : y^2 + xy = x^3 - 2x + 1$ for which the discriminant $\Delta = -61$

1. By computing the cardinality of $\overline{E}(\mathbb{F}_p)$ for $p = 2, 3, 5, 7$, show that $E(\mathbb{Q})_{\text{tor}}$ is trivial.
2. Prove that $E(\mathbb{Q}_2)_{\text{tor}}$ has order dividing 8
3. If $P = (1, 0) \in E(\mathbb{Q})$, prove that $7P$ and $9P$ don't have integral coordinates.

Proof. (1) By computing we have $|\overline{E}(\mathbb{F}_2)| = 4$, $|\overline{E}(\mathbb{F}_3)| = 6$, $|\overline{E}(\mathbb{F}_5)| = 9$, $|\overline{E}(\mathbb{F}_7)| = 7$. Now since $\Delta = -41$, E has good reduction at $p = 2, 3, 5, 7$. By lemma 12.2, we have $|E(\mathbb{Q})_{\text{tor}}| \mid 9 \cdot 5^a$ and $|E(\mathbb{Q})_{\text{tor}}| \mid 7 \cdot 7^b$ for some $a, b \geq 0$. Hence we have $E(\mathbb{Q})_{\text{tor}} = 0$

(2) By theorem 11.1, $E_r(K) \cong (O_K, +)$. Then for $K = \mathbb{Q}_2$, we can take $r = 2$. Also since E has good reduction at 2, $E_0(\mathbb{Q}_2) = E(\mathbb{Q}_2)$. So $E_2(\mathbb{Q}_2) \cong \mathbb{Z}_2, +$ is torsionfree, and hence $E(\mathbb{Q}_2)_{\text{tor}} \cap E_2(\mathbb{Q}_2)$ injects into $E(\mathbb{Q}_2)/E_2(\mathbb{Q}_2)$, which is a finite group of order 8 since $|E(\mathbb{Q}_2)/E_1(\mathbb{Q}_2)| = |\overline{E}(\mathbb{F}_2)| = 4$ and $E_1(\mathbb{Q}_2)/E_2(\mathbb{Q}_2) \cong \mathbb{F}_2, +$.

(3) First $7P, 9P \neq 0$ by (1). Notice that $7\overline{P} = 7\overline{P} = 0 \in \overline{E}(\mathbb{F}_7)$ and $9\overline{P} = 9\overline{P} = 0 \in \overline{E}(\mathbb{F}_5)$. So by definition $7P \in E_1(\mathbb{Q}_7)$, and so the coordinate of $7P$ is not in $\mathbb{Z}_7$ and hence not in $\mathbb{Z}$. Same for $9P$. $\square$

Proposition 12.2. Let $E/\mathbb{Q}$ be the elliptic curve $y^2 = x^3 + \lambda x$ where λ is a (non-zero) integer. For p a prime not dividing 2λ write $|E(\mathbb{F}_p)| = p + 1 - a_p$. Show that if $p = 4k + 1$ then

$$a_p \equiv \binom{2k}{k} \lambda^k \pmod{p}$$

Deduce that $a_p \equiv 0 \pmod{p}$ if and only if $p \equiv 3 \pmod{4}$

Proof. Recall that we have $|E(\mathbb{F}_p)| = p + 1 + \sum_{x=0}^{p-1} \left(\frac{x^3 + \lambda x}{p} \right)$. Hence $a_p = -\sum_{x=0}^{p-1} \left(\frac{x^3 + \lambda x}{p} \right) = -\sum_{x=1}^{p-1} \left(\frac{x + \lambda/x}{p} \right)$. Now for $p = 4k + 1$, we have

$$a_p = -\sum_{x=1}^{p-1} (x + \lambda/x)^{\frac{p-1}{2}} = -\sum_{x=1}^{p-1} (x + \lambda/x)^{2k}$$

Note that we also have $\sum_{x \in \mathbb{F}_p^\times} x^r \equiv p-1 \equiv -1 \pmod{p}$ for $p-1 \mid r$ and $\sum_{x \in \mathbb{F}_p^\times} x^r \equiv 0 \pmod{p}$ else, which can be seen by writing them as powers of primitive root. For this reason we see that in the expansion of $-\sum_{x=1}^{p-1} (x + \lambda/x)^{2k}$ only the term where x gets cancelled survives. So $a_p \equiv -(p-1) \binom{2k}{k} \lambda^k \equiv \binom{2k}{k} \lambda^k \pmod{p}$

Now it is enough to show that if $p \equiv 3 \pmod{4}$ then $a_p \equiv 0 \pmod{p}$. This is clear once we noticed that $x^3 + \lambda x$ and $(\frac{-1}{p}) = -1$. So we can pair them up to cancel each other. $\square$

Proposition 12.3. Show that for $E : y^2 = x^3 + ax^2 + bx$ with $a, b \in \mathbb{Z}$ and $P = (x, y) \in E(\mathbb{Q})$ is a torsion point. Then either $x = 0$ or x divides b and $x + a + b/x$ is a perfect square

Proof. For elliptic curves of this form, we know that we have the 2-isogeny $\phi : E \rightarrow E'$ given by $(x, y) \mapsto ((\frac{y}{x})^2, \frac{y}{x^2}(x^2 - b))$, where $E'/K : y^2 : x(x^2 + a'x + b')$ with $a' = -2a$ and $b' = a^2 - 4b$. If P is a torsion point then so is $\phi(P) = ((\frac{y}{x})^2 = \frac{x^3 + ax^2 + bx}{x^2} = x + a + b/x, \frac{y}{x^2}(x^2 - b))$. Now suppose $x \neq 0$. Since $a', b' \in \mathbb{Z}$, by Nagell-Lutz we have x and $x + a + b/x$ are integers. Hence $x \mid b$. Also $(\frac{y}{x})^2 = x + a + b/x$ is a integer implies that y/x is an integer and $x + a + b/x$ is a perfect square. $\square$

Proposition 12.4. *Let d be a positive integer and let $E/\mathbb{Q}$ be the elliptic curve $y^2 = x^3 + d$. Show that if $p \nmid 6d$ is a prime congruent to $2 \pmod{3}$ then $|\overline{E}(\mathbb{F}_p)| = p + 1$. Show that $E(\mathbb{Q})_{\text{tor}}$ has order dividing 6, and the elliptic curves has infinitely many $\mathbb{Q}$ -points.*

Proof. Since $p - 1 \equiv 1 \pmod{3}$, we can show that x^3 passes through $\mathbb{F}_p^\times$ when x passes through $\mathbb{F}_p^\times$. Then $|E(\mathbb{F}_p)| = p + 1 + \sum_{x=0}^{p-1} (\frac{x^3+d}{p}) = p + 1 + \sum_{x \in \mathbb{F}_p} (\frac{x}{p}) = p + 1$

Now suppose some prime $l \neq 2, 3$ divides $|E(\mathbb{Q})_{\text{tor}}|$. Then for any p with in the settings with $p \nmid 6dl$, we have $l \mid |\overline{E}(\mathbb{F}_p)| = p + 1$ by (1). Then $p \equiv -1 \pmod{l}$. This is a contradiction to Dirichlet's theorem on primes in arithmetic progressions. Similarly, $4, 9 \nmid |E(\mathbb{Q})_{\text{tor}}|$. So we are done.

For $E : y^2 = x^3 + 5$, we compute that $|\overline{E}(\mathbb{F}_7)| = 7$. Hence $|E(\mathbb{Q})_{\text{tor}}| \mid 7^{a+1}$ for some $a \geq 0$ as E has good reduction at 7. With (2), we get $E(\mathbb{Q})_{\text{tor}} = 0$. Note that $P = (-1, 2) \in E(\mathbb{Q})$. Since there is no nontrivial torsion point, P is torsionfree, and hence generate infinitely many $\mathbb{Q}$ -points on E $\square$

Proposition 12.5. (i) *Let E be an elliptic curve over an algebraically closed field K . Let $\phi : E \rightarrow E$ be a morphism of curves (not necessarily an isogeny). Show that if ϕ has no fixed points, then ϕ (and hence also ϕ^n) is a translation map.*

(ii) *Let $C/\mathbb{F}_q$ be a smooth projective curve of genus one. Show that $C(\mathbb{F}_q) \neq \emptyset$.*

Proof. (i) Suppose ϕ has no fixed points. Then $\phi - id$ is constant, since otherwise it is surjective and the preimage of $\mathcal{O}_E$ is a fixed point. Thus there is $Q \in E(K)$ such that $\phi(P) - P = Q$ for all $P \in E(K)$ as required.

(ii) Let $C \subset \mathbb{P}^n$. Since C is defined over $\mathbb{F}_q$, we have the q -th power Frobenius on C . Now take $P = (a_0, \dots, a_n) \in C(\overline{\mathbb{F}_q})$, then there is $d \in \mathbb{N}$ such that $P \in C(\mathbb{F}_{q^d})$. So ϕ^d fixes P .

Now suppose $C(\mathbb{F}_q) = \emptyset$. Note that (C, P) is an elliptic curve. Since $C(\mathbb{F}_q) = \emptyset$, ϕ has no fixed point. By (i), ϕ and hence ϕ^d is a translation. But ϕ^d fixes P implies that ϕ^d is identity. But this implies $C(\overline{\mathbb{F}_q}) = C(\mathbb{F}_{q^d})$ is finite, contradiction. $\square$

13 Kummer's Theorem

Suppose K a field, $n \geq 1$ and $\text{char}(K) \nmid n$. Assume $\mu_n \in K$. Let $\Delta \subset K^\times / (K^\times)^n$ a finite subgroup and $L = K(\sqrt[n]{\Delta})$. We define the **Kummer pairing**: $\langle -, - \rangle : \text{Gal}(L/K) \times \Delta \rightarrow \mu_n; (\sigma, x) \mapsto \frac{\sigma(\sqrt[n]{x})}{\sqrt[n]{x}}$. This is well-defined i.e. independent of the choice of $\sqrt[n]{x}$ because $\mu_n \in K$.

Proposition 13.1. *The Kummer pairing is bilinear and non-degenerate*

Corollary 13.1. *We have isomorphisms $\text{Gal}(L/K) \rightarrow \text{Hom}(\Delta, \mu_n); \sigma \mapsto \langle \sigma, - \rangle$ and $\Delta \rightarrow \text{Hom}(\text{Gal}(L/K), \mu_n); x \mapsto \langle -, x \rangle$*

Proof. Since $\langle -, - \rangle$ is non-degenerate, these two maps are injective. Now recall that for a finite abelian group G of exponent n , $\text{Hom}(G, \mu_n) \cong G$. Since $\Delta \subset K^\times / (K^\times)^n$ is of exponent n , $\text{Hom}(\Delta, \mu_n) \cong \Delta$. Moreover, the first map identifies $\text{Gal}(L/K)$ as a subgroup of Δ and hence $\text{Gal}(L/K)$ is of exponent n . So $\text{Hom}(\text{Gal}(L/K), \mu_n) \cong \text{Gal}(L/K)$. So these maps are isomorphisms for cardinality reason. $\square$

In the proof we showed that $\text{Gal}(L/K)$ is a **finite abelian extension of exponent n** . This is precisely the content of Kummer's theorem

Theorem 13.1. (Kummer theory) *There is a bijection between finite subgroup $\Delta \subset K^\times / (K^\times)^n$ and finite abelian extensions of exponent n over K , given by $\Delta \mapsto K(\sqrt[n]{\Delta})$*

For a finite abelian extension of exponent n L/K , define $\Delta = K^\times \cap (L^\times)^n / (K^\times)^n$. This gives an inverse.

Now take K a number field

Definition 13.1. *The n -Selmer group of K is $Sel_n(K) = \{x \in K^\times / (K^\times)^n \mid v_{\mathfrak{p}}(x) \equiv 0 \pmod n \forall \text{ prime } \mathfrak{p}\}$*

Lemma 13.1. *We have a s.e.s $0 \rightarrow \mathcal{O}_K^\times / (\mathcal{O}_K^\times)^n \xrightarrow{i} Sel_n(K) \xrightarrow{\Pi} Cl(K)[n] \rightarrow 0$. This implies that $Sel_n(K)$ is finite.*

We will only define the maps. i is induced by the inclusion $\mathcal{O}_K^\times \hookrightarrow K^\times$. For $x \in K^\times$ represents a class $[x]$ in $Sel_n(K)$, we have $(x) = \mathfrak{a}^n$ for some fractional ideal $\mathfrak{a}$. We define $\Pi([x]) = [\mathfrak{a}] \in Cl(K)[n]$. Now since $\mathcal{O}_K^\times / (\mathcal{O}_K^\times)^n$ is finite by Dirichlet's unit theorem and $Cl(K)[n]$ is finite, $Sel_n(K)$ is finite.

Proposition 13.2. *Let K be a number field with $\mu_n \subset K$ and S a finite set of places of K . There are only finitely many extensions L/K such that L/K is abelian of exponent dividing n , and L/K is unramified at all $\mathfrak{p} \notin S$*

Proof. By Kummer theory, any such extension L/K is of the form $K(\sqrt[n]{\Delta})$ for some finite subgroup $\Delta \subset K^\times / (K^\times)^n$. Suppose $\mathfrak{p} \notin S$ prime and $x \in \Delta$, $K(\sqrt[n]{x})/K$ unramified at $\mathfrak{p}$ implies that for any prime $\mathfrak{P} \mid \mathfrak{p}$ of $K(\sqrt[n]{x})$, $v_{\mathfrak{P}}(x) = v_{\mathfrak{p}}(x) = nv_{\mathfrak{P}}(\sqrt[n]{x}) \equiv 0 \pmod n$. Thus $\Delta \subset K(S, n) = \{x \in K^\times / (K^\times)^n \mid v_{\mathfrak{p}}(x) \equiv 0 \pmod n \forall \mathfrak{p} \notin S\}$

Now $K(S, n) \rightarrow (\mathbb{Z}/n\mathbb{Z})^{|S|}; [x] \mapsto (v_{\mathfrak{p}}(x))_{\mathfrak{p} \in S}$ is a homomorphism with kernel $Sel_n(K)$ which is finite by lemma 13.1. Since $(\mathbb{Z}/n\mathbb{Z})^{|S|}$ is finite, we get $K(S, n)$ is finite. So there are only finite possibilities of Δ as a subgroup. $\square$

14 Weak Mordell-Wéil

In this section we want to prove the **Weak Mordell-Wéil theorem**

Theorem 14.1. *K number field, E/K an elliptic curve, $n \geq 2$. Then $E(K)/nE(K)$ is finite.*

We start with a special case

Proposition 14.1. *Suppose $\mu_n, E[n] \subset K$, then $E(K)/nE(K)$ is finite.*

Proof. Let $P \in E(K)$ and $L = K([n]^{-1}P) = K(\frac{1}{n}P)$ Galois over K , which is independent of the choice of $\frac{1}{n}P$ since $E[n] \subset K$. Take any Q with $nQ = P$ and define a homomorphism $Gal(L/K) \rightarrow E[n]; \sigma \mapsto \sigma(Q) - Q$. This is injective because the coords of Q generate L . Thus $Gal(L/K) \subset E[n] \cong (\mathbb{Z}/n\mathbb{Z})^2$, so L/K is abelian with exponent n . Moreover, by theorem 11.3, L/K is unramified over $\mathfrak{p}$ with good reduction, i.e. outside the set of places $S = \{\mathfrak{p} \text{ prime} \mid E \text{ has bad reduction at } \mathfrak{p}\}$, which is finite by lemma 12.1. By proposition 13.2, there are only finite possibilities for $K(\frac{1}{n}P)/K$.

We take the compositum F of all such extensions. Hence F is still finite Galois over K with exponent n . Define the "elliptic curve Kummer pairing" $\langle -, - \rangle : Gal(F/K) \times E(K)/nE(K) \rightarrow E[n]; (\sigma, P) \mapsto \sigma(Q) - Q$ for any $nQ = P$. This is well-defined because $E[n] \subset E(K)$. This is bilinear and has trivial right kernel, and hence induces an injection $E(K)/nE(K) \hookrightarrow Hom(Gal(F/K), E[n])$, and the latter group is finite. $\square$

The key point is to show that the compositum of all $K([n]^{-1}P)$ is a finite extension since they are all abelian of exponent n and unramified outside S .

Lemma 14.1. *K number field, L/K finite Galois, E/K elliptic curves. Then the natural map $E(K)/nE(K) \rightarrow E(L)/nE(L)$ has finite kernel.*

Proof. Fix coset representatives $\{P_i\}$ for the kernel. For each i , choose $Q_i \in E(L)$ such that $nQ_i = P_i$ since $P_i \in nE(L)$. Define function $\alpha_i : \text{Gal}(L/K) \rightarrow E[n]; \sigma \mapsto \sigma(Q_i) - Q_i$ which is in general not a homomorphism when $E[n] \not\subset K$. If $\alpha_i = \alpha_j$, then $\sigma(Q_i - Q_j) = Q_i - Q_j$ for all $\sigma \in \text{Gal}(L/K)$ and hence $Q_i - Q_j \in E(K)$ and $P_i - P_j \in nE(K)$ and so $i = j$. Thus we have an injection from the kernel of $E(K)/nE(K) \rightarrow E(L)/nE(L)$ to the set of functions $\text{Gal}(L/K) \rightarrow E[n]$, which is finite. $\square$

Proof of Weak Mordell-Wéil. Let $L = K(E[n], \mu_n)$ which is finite Galois over K . Then $E(L)/nE(L)$ is finite by proposition 14.1. Hence $E(K)/nE(K)$ is finite by lemma 14.1 $\square$

Remark 14.1. We will see that $\mu_n \subset K(E[n])$. Galois cohomology can help simplify the proof of lemma 14.1

14.1 Exercise

Proposition 14.2. Let $[K : \mathbb{Q}_p] < \infty$. Let E/K an elliptic curve and n an integer. Show that

$$|E(K)/nE(K)| = |E(K)[n]| \cdot |\mathcal{O}_K/n\mathcal{O}_K|$$

To prove this we need the following lemma:

Lemma 14.2. Suppose $A \subset B$ is a subgroup of finite index. Define $q(A) = \text{coker}(A \xrightarrow{[n]} A) / \text{ker}(A \xrightarrow{[n]} A)$. Then $q(A) = q(B)$

This can be seen by applying snake's lemma on the commutative diagram induced by $[n]$

Proof of proposition. For K a local field, for $r > \frac{e}{p-1}$ we have $E_r(K) \cong \mathcal{O}_K, +$ and is of finite index in $E(K)$ by theorem 11.2. So we apply lemma 14.2 to $A = E_r(K) \cong \mathcal{O}_K$ and $B = E(K)$.

Note that $n = u\pi^{v(n)}$ and $\mathcal{O}_K/n\mathcal{O}_K = \mathcal{O}_K/\pi^{v(n)}\mathcal{O}_K$. It is a $k = \mathcal{O}_K/m$ -vector space of dimension $v(n)$ since m^n/m^{n+1} is a dim-1 k -vector space. So $|\mathcal{O}_K/\pi^{v(n)}\mathcal{O}_K| = \#|k|^{v(n)}$ $\square$

Remark 14.2. We can use Mordell-Wéil to drive similar results over number fields.

15 Heights

Definition 15.1. Given $x \in \mathbb{Q}$, define the height $H(x) = \max\{|a|, |b|\}$, where $x = a/b$ with $a, b \in \mathbb{Z}$ coprime, and $H(0) = 1$

More generally, for $P \in \mathbb{P}^n(\mathbb{Q})$, write $P = [a_0 : \dots : a_n]$ with $a_i \in \mathbb{Z}$, $\gcd(a_0, \dots, a_n) = 1$. Then $H(P) = \max\{|a_i|\}$ and hence $H(x) = H([x : 1])$ for $x \in \mathbb{Q}$

For $B \in \mathbb{R}_{>0}$, note that the set $\{P \in \mathbb{P}^n(\mathbb{Q}) \mid H(P) \leq B\}$ is finite

We will omit or give a sketch of most proofs in this section.

Proposition 15.1. Let $\Phi : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ a morphism of degree d defined over $\mathbb{Q}$. Then there is $c_1, c_2 > 0$ such that for all $P \in \mathbb{P}^1(\mathbb{Q})$, $c_1 H^d(P) \leq H(\Phi(P)) \leq c_2 H^d(P)$

Now given $E/\mathbb{Q} : y^2 = x^3 + ax + b$ an elliptic curve,

Definition 15.2. The **height** function is $H : E(\mathbb{Q}) \rightarrow \mathbb{R}_{\geq 0}$ given by $P \mapsto H(x(P))$ if $P = (x(P), y(P))$ and 1 if $P = \mathcal{O}_E$. The **logarithm height** $h : E(\mathbb{Q}) \rightarrow \mathbb{R}_{\geq 0}$ is given by $P \mapsto \log H(P)$

Lemma 15.1. Given E, E' two elliptic curves over $\mathbb{Q}$ and $\phi : E \rightarrow E'$ an isogeny defined over $\mathbb{Q}$. Then there is a constant c depending on E, E' and ϕ such that $|h(\phi(P)) - \deg(\phi)h(P)| < c$ for all $P \in E(\mathbb{Q})$

Indeed we have a commutative diagram

$$\begin{array}{ccc} E(\mathbb{Q}) & \xrightarrow{\phi} & E'(\mathbb{Q}) \\ \downarrow x & & \downarrow x \\ \mathbb{P}^1 & \xrightarrow{\Phi} & \mathbb{P}^1 \end{array}$$

with $\deg(\phi) = \deg(\Phi)$. Hence $H(\phi(P)) =$

$H(x(\phi(P))) = H(\Phi(x(P)))$. By taking logarithm, this follows from proposition 15.1

Remark 15.1. In particular there is a constant c such that $|h(2P) - 4h(P)| < c$ for all $P \in E(\mathbb{Q})$

Definition 15.3. The **canonical height** $\hat{h} : E(\mathbb{Q}) \rightarrow \mathbb{R}_{\geq 0}$ is defined by $\hat{h}(P) = \lim_{n \rightarrow \infty} \frac{1}{4^n} h(2^n P)$

The limit in this definition exists because for $m \geq n$, $|\frac{1}{4^m} h(2^m P) - \frac{1}{4^n} h(2^n P)| \leq \sum_{r=n}^{m-1} |\frac{1}{4^{r+1}} h(2^{r+1} P) - \frac{1}{4^r} h(2^r P)| < \sum_{r=n}^{m-1} \frac{1}{4^{r+1}} c < \frac{1}{3 \cdot 4^n} c \rightarrow 0$ when $n \rightarrow \infty$ by remark 15.1. Moreover, by letting $n = 0$ and $m \rightarrow \infty$, we see that $|\hat{h}(P) - h(P)|$ is bounded by a constant $\frac{c}{3}$ for all $P \in E(\mathbb{Q})$.

Corollary 15.1. For all $B \in \mathbb{R}_{\geq 0}$, then set $\{P \in E(\mathbb{Q}) | \hat{h}(P) \leq B\}$ is finite.

This follows from the above observations

Corollary 15.2. For all isogenies $\phi : E \rightarrow E'$, $\hat{h}(\phi(P)) = \deg(\phi) \hat{h}(P)$

By lemma 15.1, $|h(\phi(P)) - \deg(\phi)h(P)| < c$ for some constant c and all $P \in E(\mathbb{Q})$. Replace P with $2^n P$ and divide by 4^n and take $n \rightarrow \infty$, we get this result.

Corollary 15.3. $P \in E(\mathbb{Q})_{\text{tor}}$ if and only if $\hat{h}(P) = 0$

If $nP = 0$, then $0 = \hat{h}(nP) = n^2 \hat{h}(P) \implies \hat{h}(P) = 0$ by corollary 15.2. Conversely, if $\hat{h}(P) = 0$, then the set $\{P, 2P, \dots\}$ is of set of points of height 0. By corollary 15.1, this set is finite and hence P is torsion.

Corollary 15.4. $\hat{h}$ is independent of the choice of Weierstrass equations for E

Take ϕ an isomorphism in corollary 15.2

Theorem 15.1. $\hat{h} : E(\mathbb{Q}) \rightarrow \mathbb{R}_{\geq 0}$ is a quadratic form

We need to use the following lemma on height function:

Lemma 15.2. There is c constant such that $H(P+Q)H(P-Q) \leq cH(P)^2H(Q)^2$ for all $P, Q \in E(\mathbb{Q})$ with $P, Q, P+Q \neq 0$.

Now using this lemma, take logarithm and we get $h(P+Q) + h(P-Q) \leq 2h(P) + 2h(Q) + \text{some constant}$ for all $P, Q \in E(\mathbb{Q})$. Indeed, for $P, Q, P+Q = 0$ this follows from $|h(2P) - 4h(P)|$ is bounded. Now replace P, Q with $2^n P, 2^n Q$ and divide by 4^n and take $n \rightarrow \infty$, we get $\hat{h}(P+Q) + \hat{h}(P-Q) \leq 2\hat{h}(P) + 2\hat{h}(Q)$. Replace P, Q with $P-Q, P+Q$, we get the other direction. So $\hat{h}(P+Q) + \hat{h}(P-Q) = 2\hat{h}(P) + 2\hat{h}(Q)$. So $\hat{h}$ satisfies the parallelogram law and hence is a quadratic form.

We can generalize these results to number fields. Let Σ_K the **set of normalized absolute values** of K , consisting of finite places (corresponding to primes $\mathfrak{p}$ of K) and infinite places (corresponding to real and complex embeddings of K into $\mathbb{C}$). For any $x \in K^\times$, we have the product formula

$$\prod_{v \in \Sigma_K} |x|_v = 1$$

Given $P = [a_0 : \dots : a_n] \in \mathbb{P}^n(K)$, define the height function $H(P) = \prod_{v \in \Sigma_K} \max\{|a_i|_v\}$. This is well-defined because of the product formula. All results in this section can generalize from $\mathbb{Q}$ to K using this height function. In particular, we still have the canonical height $\hat{h}$ satisfying corollary 15.2, 15.3, 15.4 and is a quadratic form by theorem 15.1

Theorem 15.2. (Mordell-Wéil) E/K an elliptic curve over a number field K . Then $E(K)$ is a finitely generated abelian group.

Proof. Fix any $n \geq 2$. The weak Mordell-Wéil says that $E(K)/nE(K)$ is finite. Take coset representatives $P_1, \dots, P_k$. We claim that the set $S = \{P \in E(K) | \hat{h}(P) \leq \max\{\hat{h}(P_i)\}\}$ which is finite by corollary 15.1 generates $E(K)$.

If not, let $T = E(K) \setminus \langle S \rangle$. Take $P \in T$ with $\hat{h}(P)$ minimal, which is possible by corollary 15.1. Suppose $[P] = [P_i]$ in $E(K)/nE(K)$. Then $P = P_i + nQ$ for some $Q \in E(K)$. Note that $P \notin S \implies Q \notin S$. So $Q \in T$. Then $\hat{h}(P - P_i) = \hat{h}(nQ) = n^2\hat{h}(Q) \geq 4\hat{h}(Q) \geq 4\hat{h}(P)$ by the minimality of P . But $\hat{h}(P - P_i) \leq \hat{h}(P - P_i) + \hat{h}(P + P_i) = 2\hat{h}(P) + 2\hat{h}(P_i)$ since $\hat{h}$ is a quadratic form. So $\hat{h}(P) \leq \hat{h}(P_i)$ and so $P \in S$, a contradiction. $\square$

Now by structure theorem for f.g. abelian groups, $E(K) = E(K)_{\text{tor}} \times \mathbb{Z}^r$. The integer r is the **rank** of E , denoted by $\text{rank}(E)$

15.1 Exercise

Proposition 15.2. Let $E/\mathbb{Q} : y^2 = x^3 + ax + b$ be an elliptic curve and $E_d : dy^2 = x^3 + ax + b$ its quadratic twist by a square free integer d . Let $K = \mathbb{Q}(\sqrt{d})$. Show that $\text{rank}(E(K)) = \text{rank}(E(\mathbb{Q})) + \text{rank}(E_d(\mathbb{Q}))$

Proof. First note that E_d and E are isomorphic over K by $\phi : E_d \rightarrow E; (x, y) \mapsto (x, \sqrt{d}y)$. Now we define a homomorphism $\alpha : E(\mathbb{Q}) \times E_d(\mathbb{Q}) \rightarrow E(K)$ given by $\alpha(P, Q) = P - \phi(Q)$. It is enough to show that α has finite kernel and cokernel.

Now let $G = \text{Gal}(K/\mathbb{Q})$ with generator σ . Then we have $E(\mathbb{Q}) = \{P \in E(K) | \sigma P = P\}$. Also we can check that $\phi(E_d(\mathbb{Q})) = \{P \in E(K) | \sigma P = -P\}$. Note that for any $P \in E(K)$, $2P = P + \sigma P + P - \sigma P$. Note that $P + \sigma P$ is fixed by σ and hence in $E(\mathbb{Q})$. Also $P - \sigma P$ is inverted by σ and hence in $\phi(E_d(\mathbb{Q}))$. Hence $2E(K) \subset \text{im}(\alpha)$. Hence $|\text{coker}(\alpha)| < |E(K)/2E(K)|$ which is finite.

Let $(P, Q) \in \ker \alpha$. Then $P = \phi(Q)$. Apply σ and we get $P = -\phi(Q)$. Hence $2P = \mathcal{O}_E$ and $2\phi(Q) = \phi(2Q) = \mathcal{O}_E \implies 2Q = \mathcal{O}_{E_d}$ since ϕ is an isomorphism over K . So $\ker \alpha \subset E(\mathbb{Q})[2] \times E_d(\mathbb{Q})[2]$ is finite. (Or just notice that the y -coordinate of P is rational while $\phi(Q)$'s is not) $\square$

16 Dual Isogenies and the Wéil Pairing

Let $K = \overline{K}$, $\phi : E \rightarrow E'$ an isogeny, $\phi^* : K(E') \hookrightarrow K(E)$, $\deg(\phi) = [K(E) : \phi^*K(E')]$

Lemma 16.1. Suppose ϕ is separable. $E[\phi] = \ker \phi$ has order $\deg(\phi)$

Because ϕ is separable, we know that $\phi^{-1}Q$ has order $\deg \phi$ for all but finitely many $Q \in E'$. Also note that there is a bijection $E[\phi] \rightarrow \phi^{-1}Q; T \mapsto T + R$ for any $Q \in E'$ $R \in \phi^{-1}Q$.

Corollary 16.1. Suppose ϕ is separable, then ϕ is unramified, i.e. $e_\phi(P) = 1$ for all $P \in E$

This is by the formula $\deg \phi = \sum_{P \in \phi^{-1}Q} e_\phi(P)$

Lemma 16.2. Suppose ϕ is separable, then $K(E)/\phi^*(K(E'))$ is Galois and $E[\phi] \cong \text{Gal}(K(E)/\phi^*(K(E')))$ given by $P \mapsto T_P^*$

Proof. Since $P \in E(\phi)$, $T_P^* \circ \phi^* = (\phi \circ T_P)^* = \phi^*$. So T_P^* acts trivially on $\phi^*(K(E'))$. So this is well-defined.

Since $T_{P+Q} = T_P + T_Q$, this is homomorphism. Moreover, if T_P^* acts trivially on $K(E)$, then $T_P^* \circ x = x \circ T_P = x$. Since x has a pole at $\mathcal{O}_E$, we see x has a pole at P as well. But $\mathcal{O}_E$ is the only pole of x , so $P = \mathcal{O}_E$. This shows that this map is injective. Now by lemma 6.1, this is an isomorphism for cardinality reason. $\square$

Proposition 16.1. *K a perfect field and E/K an elliptic curve. Let $N \subset E(\overline{K})$ a finite subgroup stable under Γ_K . Then there exists an elliptic curve E'/K and a separable isogeny $\phi : E \rightarrow E'$ defined over K such that $E[\phi] = N$, and every isogeny $\phi' : E \rightarrow E''$ with $N \subset E[\phi']$ uniquely factors through ϕ .*

In particular, for a separable ϕ with $\deg \phi = n$, we see that $E[\phi]$ is an abelian group of order n and hence killed by n . So $E[\phi] \subset E[n]$. Hence by proposition 16.1, $[n] : E \rightarrow E$ uniquely factors through ϕ , i.e. there is a unique isogeny $\hat{\phi} : E' \rightarrow E$ such that $\hat{\phi} \circ \phi = [n]$. $\hat{\phi}$ is the **dual isogeny** of ϕ .

Example 8. (2-isogeny) Suppose $\text{char}(K) \neq 2$, $0 \neq T \in E(K)[2]$. WLOG we assume $E : y^2 = x(x^2 + ax + b)$ and $T = (0, 0)$. If $P = (x, y) \in E(K)$. Then by explicit formula, $\mathcal{T}_T^*(P) = P + T = ((\frac{y}{x})^2 - a - x, \frac{-by}{x^2})$. Define $u = x + \mathcal{T}_T^*(x) + a = (\frac{y}{x})^2$ and $v = y + \mathcal{T}_P^*(y) = \frac{y}{x^2}(x^2 - b)$. One can check that $K(u, v) = K(E)^G$ where $G = \langle 1, \mathcal{T}_T^* \rangle$.

We compute that $v^2 = u((u - a)^2 - 4b)$. So we define $E' : y^2 = x(x^2 + a'x^2 + b'x)$ with $a' = -2a$ and $b' = a^2 - 4b$ and an isogeny $\phi : E \rightarrow E'$, $(x, y) \mapsto ((\frac{y}{x})^2, \frac{y}{x^2}(x^2 - b))$. Then by $E \xrightarrow{x} \mathbb{P}^1$ and the omitted theorem, $\deg \phi = 2$ and $T \in \ker \phi$. So $\ker \phi = \{0, T\}$

Lemma 16.3. *Let $\phi \in \text{End}(E)$, then $[\text{tr} \phi] = \phi + \hat{\phi}$*

Proof. Recall that $\phi^2 - [\text{tr} \phi]\phi + [\deg \phi] = 0$. So $[\deg \phi] = ([\text{tr} \phi] - \phi) \circ \phi$. By uniqueness of dual isogeny we see that $[\text{tr} \phi] - \phi = \hat{\phi}$ $\square$

Corollary 16.2. $\widehat{\phi + \varphi} = \hat{\phi} + \hat{\varphi}$ for $\phi, \varphi \in \text{Hom}(E_1, E_2)$

If $E_1 = E_2$, this follows from lemma 16.3 and linearity of trace. In general, we can reduce to this case by composition with any isogeny $\alpha \in \text{Hom}(E_2, E_1)$

Recall that we have the commutative diagram

$$\begin{array}{ccc} E & \xrightarrow{\phi} & E' \\ \downarrow \alpha_1, \sim & & \downarrow \alpha_2, \sim \\ Cl^0(E) & \xrightarrow{\phi_*} & Cl^0(E') \end{array}$$

For $\hat{\phi}$, we also have the

following commutative diagram

$$\begin{array}{ccc} E' & \xrightarrow{\hat{\phi}} & E \\ \downarrow \alpha_2, \sim & & \downarrow \alpha_1, \sim \\ Cl^0(E') & \xrightarrow{\hat{\phi}^*} & Cl^0(E) \end{array}$$

In fact, this is one way to define $\hat{\phi}$.

Our next aim is to define Wéil pairing. Suppose $K = \overline{K}$, $n \geq 1$ and $\text{char}(K) \nmid n$ and $\phi : E \rightarrow E'$ an isogeny of degree n . $\text{char}(K) \nmid n$ implies $[n]$ is separable.

By lemma 16.2, $K(E)/\phi^*(K(E'))$ is Galois and $E[\phi] \cong \text{Gal}(K(E)/\phi^*(K(E')))$, which is abelian and of exponent n since $\deg \phi = n$. Hence Kummer's theorem shows that $K(E')^\times \cap (K(E)^\times)^n / (K(E')^\times)^n$ is a finite subgroup of $K(E')^\times / (K(E')^\times)^n$. Note that $\mu_n \subset K$. So we have the Kummer pairing

$$E[\phi] \cong \text{Gal}(K(E)/\phi^*(K(E'))) \times \frac{K(E')^\times \cap (K(E)^\times)^n}{(K(E')^\times)^n} \rightarrow \mu_n$$

Now for $Q \in E'[\hat{\phi}]$, $\deg \hat{\phi} = n$ implies that $E'[\hat{\phi}]$ is n -torsion, $nQ = 0$. Then $E' \xrightarrow{\alpha', \sim} Cl(E')$; $P \mapsto [(P) - (\mathcal{O}_E)]$ implies that $0 = \alpha'(nQ) = [n(Q) - n(\mathcal{O}_E)]$. So $n(Q) - n(\mathcal{O}_E) = \text{div} f_Q$ for some $f_Q \in K(E')^\times$

Lemma 16.4. $Q \mapsto f_Q$ gives an isomorphism $E'[\hat{\phi}] \xrightarrow{\sim} \frac{K(E')^\times \cap (K(E)^\times)^n}{(K(E')^\times)^n}$

Proof. First we show that it is well-defined. Follow the diagram above, $Q \in E'[\widehat{\phi}]$ implies that $[\phi^*((Q) - (\mathcal{O}_E))] = 0$. So there is $g \in K(E)^\times$ such that $\text{div}(g) = \phi^*((Q) - (\mathcal{O}_E))$. So $\text{div}(g^n) = n\phi^*((Q) - (\mathcal{O}_E)) = \phi^*(n(Q) - n(\mathcal{O}_E)) = \phi^*\text{div}(f) = \text{div}(\phi^*f) \implies g^n = \lambda\phi^*f$ for some constant $\lambda \in K$. Since $K = \overline{K}$, we see that $\phi^*f \in (K(E)^\times)^n$, and so this map actually lands in $\frac{K(E')^\times \cap (K(E)^\times)^n}{(K(E')^\times)^n}$.

Now for $P, Q \in E'[\widehat{\phi}]$, α' is an isomorphism implies that $(P + Q) - (\mathcal{O}_E) \sim (P) + (Q) - 2(\mathcal{O}_E) \implies (P + Q) - (\mathcal{O}_E) - ((P) + (Q) - 2(\mathcal{O}_E)) = \text{div}(g)$ for some $g \in K(E')^\times \implies \text{div}(f_{P+Q}) - \text{div}(f_P) - \text{div}(f_Q) = \text{div}(g^n) \implies f_{P+Q}/(f_P f_Q) = \lambda g^n = 1$ when modulo $(K(E')^\times)^n$. So it is a homomorphism.

If $f_Q = g^n$ for some $g \in K(E')$, then $n(Q) - n(\mathcal{O}_{E'}) = \text{div}f_Q = \text{div}(g^n) \implies [(Q) - (\mathcal{O}_E)] = 0 \implies Q = \mathcal{O}_E$ since α' is an isomorphism.

Finally, $E'[\widehat{\phi}]$ is of order $\deg\phi = n$ by lemma 16.1. We also has an isomorphism $\frac{K(E')^\times \cap (K(E)^\times)^n}{(K(E')^\times)^n} \cong \text{Hom}(E[\phi], \mu_n) \cong E[\phi]$ from Kummer pairing by corollary 13.1. So $\frac{K(E')^\times \cap (K(E)^\times)^n}{(K(E')^\times)^n}$ has order $\deg\phi = n$. Since this map is injective, it is an isomorphism by counting. $\square$

Now we define the **Wéil pairing**

$$e_\phi : E[\phi] \times E'[\widehat{\phi}] \rightarrow \mu_n; (P, Q) \mapsto \langle \mathcal{T}_P^*, f_Q \rangle$$

So Wéil pairing is constructed from Kummer pairing with two additional isomorphisms. Since Kummer pairing is bilinear and non-degenerate, Wéil pairing is bilinear and non-degenerate as well.

Remark 16.1. Explicitly, let $Q \in E'[\widehat{\phi}]$ and choose $g \in K(E)^\times$ such that $\phi^*((Q) - (\mathcal{O}_E)) = \text{div}(g)$ by the diagram. Since $nQ = 0$, $n(Q) - n(\mathcal{O}_E) \sim 0$ and hence $\text{div}(g^n) = \text{div}(\phi^*f)$ for some $f \in K(E')^\times$. Then for any $R \in E \setminus \text{poles or zeros of } g$, $e_\phi(P, Q) = \frac{g(R+P)}{g(R)} \in \mu_n$ since $P \in E[\phi]$

Take $\phi = [n]$, we get $e_n : E[n] \times E[n] \rightarrow \mu_n$. It lands in μ_n instead of μ_{n^2} is because $E[n]$ is n -torsion and e_n is bilinear.

Now suppose K is just a general perfect field and $\text{char}K \nmid n$

Lemma 16.5. *If E, ϕ are defined over K , then e_ϕ is Galois equivariant, i.e. $\sigma e_\phi(P, Q) = e_\phi(\sigma P, \sigma Q)$ for all $\sigma \in \Gamma_K$*

This is because for $g \in K(E)$, g is defined over K , i.e. $\sigma(g) = g$

Corollary 16.3. *If $E[n] \subset E(K)$, then $\mu_n \in K$*

Proof. Note that e_n is non-degenerate implies that there is $P, Q \in E[n]$ such that $e_n(P, Q)$ is the primitive n -th root of unity ζ . Then $\sigma(\zeta) = \zeta$ for all $\sigma \in \Gamma_K$ since $P, Q \in E(K)$. $\square$

17 Galois Cohomology

Notice if $\mu_n \not\subset K$, then for $x \in K^\times$ the function $f : \Gamma_K \rightarrow \mu_n; \sigma \mapsto \frac{\sigma(\sqrt[n]{x})}{\sqrt[n]{x}}$ depends on the choice of $\sqrt[n]{x}$, but it still satisfy the 1-cocycle condition, and given two choices of $\sqrt[n]{x}$, the resulting function differ by $\sigma \mapsto \frac{\sigma(\zeta)}{\zeta}$ for some $\zeta \in \mu_n$

Definition 17.1. *Suppose G a group and A a G -module. Define $H^0(A, G) = A^G$ the set of elements in A fixed by G . The group of 1-cochains $C_1(A, G) = \{ \text{functions } G \rightarrow A \}$. The groups of 1-cocycles $Z_1(A, G) = \{ f : G \rightarrow A \mid f(\sigma\tau) = f(\sigma) + \sigma f(\tau) \ \forall \ \sigma, \tau \in G \}$. The group of 1-coboundaries $B_1(A, G) = \{ f : G \rightarrow A \mid f(\sigma) = \sigma a - a \text{ for some } a \in A \}$ and $H^1(G, A) = Z_1(A, G)/B_1(A, G)$*

Theorem 17.1. A s.e.s of G -modules $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ induces a l.e.s

$$0 \rightarrow A^G \rightarrow B^G \rightarrow C^G \xrightarrow{\delta} H^1(G, A) \rightarrow H^1(G, B) \rightarrow H^1(G, C) \xrightarrow{\delta} \dots$$

Theorem 17.2. For a G -module A , $H \triangleleft G$ a normal subgroup, then A^H is naturally a G/H -module, and we have **inflation restriction** exact sequence

$$0 \rightarrow H^1(G/H, A^H) \xrightarrow{\text{inf}} H^1(G, A) \xrightarrow{\text{res}} H^1(H, A)$$

For K a perfect field, the absolute Galois group

$$\Gamma_K = \varprojlim_{L/K \text{ finite Galois}} \text{Gal}(L/K)$$

is a topological group.

For $G = \Gamma_K$, we modify the definition of $H^1(G, A)$ by insisting that the cochains are continuous function when A is given the discrete topology, and the stablizer of any $a \in A$ is open in G . Then

$$H^1(\Gamma_K, A) = \varinjlim_{L/K \text{ finite Galois}} H^1(\text{Gal}(L/K), A^{\text{Gal}(\bar{K}/L)})$$

Notations: We will write $H^1(K, -) := H^1(\Gamma_K, -)$

Theorem 17.3. (Hilbert's 90) For L/K finite Galois, $H^1(\text{Gal}(L/K), L^\times) = 0$

This is done by the fact that distinct automorphisms are linearly independent.

Corollary 17.1. $H^1(K, \bar{K}^\times) = 0$

Since $\bar{K}$ is perfect and $\mu_n \in \bar{K}^\times$, note that we have a s.e.s of Γ_K -modules,

$$0 \rightarrow \mu_n \rightarrow \bar{K}^\times \xrightarrow{x \mapsto x^n} \bar{K}^\times \rightarrow 0$$

By taking the Galois cohomology, we get a l.e.s

$$0 \rightarrow \mu_n(K) \rightarrow K^\times \xrightarrow{x \mapsto x^n} K^\times \xrightarrow{\delta} H^1(K, \mu_n) \rightarrow H^1(K, \bar{K}^\times) = 0$$

by Hilbert's 90. Hence we have $\frac{K^\times}{(\bar{K}^\times)^n} \cong H^1(K, \mu_n)$. If $\mu_n \in K$, then Γ_K acts trivially on μ_n , and this gives $\frac{K^\times}{(\bar{K}^\times)^n} \cong \text{Hom}_{\text{cts}}(\Gamma_K, \mu_n)$ given by $[x] \mapsto (\sigma \mapsto \frac{\sigma(\sqrt[n]{x})}{\sqrt[n]{x}})$

Lemma 17.1. $[K : \mathbb{Q}_p] < \infty$, $p \nmid n$. Then $\frac{\mathcal{O}_K^\times}{(\mathcal{O}_K^\times)^n} \cong \ker(H^1(K, \mu_n) \rightarrow H^1(K^{nr}, \mu_n))$, i.e. $\ker(H^1(K, \mu_n) \rightarrow H^1(K^{nr}, \mu_n))$ consists of those $x \in \frac{K^\times}{(\bar{K}^\times)^n}$ such that $v(x) = 0$

This weird-looking result comes from the following sequence $0 \rightarrow \frac{\mathcal{O}_K^\times}{(\mathcal{O}_K^\times)^n} \xrightarrow{\alpha} \frac{K^\times}{(\bar{K}^\times)^n} \xrightarrow{\beta} \frac{(K^{nr})^\times}{((K^{nr})^\times)^n}$ and the identifications above. So it suffices to show the exactness of this sequence.

Let $a \in \mathcal{O}_K^\times$, and $f(x) = x^n - a$. Then $\bar{f}$ has distinct roots in $\bar{k}$ since $p \nmid n$. Let k_m/k be the splitting field of $\bar{f}$ (assume to have degree m), which is the residue field extension of K_m/K (unique unramified extension of degree m). Now Hensel's lemma shows that f has a root in $K_m \subset K^{nr}$. Hence $a \in (K^{nr \times})^n$

Let $a(K^\times)^n \in \ker \beta$. Say $a = u\pi^r$ for a unit u and $r \in \mathbb{Z}$. The discrete valuation on K extends uniquely to K^{nr} without ramification, and $a = w^n$ for some $w \in K^{ur}$. Then $nv_{nr}(w) = v_{nr}(a) = v(a) =$

r. So $a(K^\times)^n = u(K^\times)^n \in \frac{\mathcal{O}_K^\times}{(\mathcal{O}_K^\times)^n}$

Back to elliptic curves. Let K a perfect field, E/K an elliptic curve and $\phi : E \rightarrow E'$ an isogeny defined over K . We have a s.e.s

$$0 \rightarrow E[\phi] \rightarrow E \xrightarrow{\phi} E' \rightarrow 0$$

The l.e.s of Galois cohomology gives $0 \rightarrow \frac{E'(K)}{\phi(E(K))} \xrightarrow{\delta} H^1(K, E[\phi]) \rightarrow H^1(K, E)[\phi] \rightarrow 0$

Take K a number field. For each place v , fix an embedding $\bar{K} \rightarrow \bar{K}_v$. We have the decomposition group $Gal(\bar{K}_v/K_v) \subset Gal(\bar{K}/K)$. Taking product of all places, we have a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \frac{E'(K)}{\phi(E(K))} & \xrightarrow{\delta} & H^1(K, E[\phi]) & \longrightarrow & H^1(K, E)[\phi] \longrightarrow 0 \\ & & \downarrow & & \downarrow \Pi res_v & \searrow \text{dashed} & \downarrow \Pi res_v \\ 0 & \longrightarrow & \prod_v \frac{E'(K_v)}{\phi(E(K_v))} & \xrightarrow{\prod_v \delta_v} & \prod_v H^1(K_v, E[\phi]) & \longrightarrow & \prod_v H^1(K_v, E)[\phi] \longrightarrow 0 \end{array}$$

Definition 17.2. The ϕ -**Selmer group** of E is $Sel_\phi(E/K)$ is $\ker\{H^1(K, E[\phi]) \rightarrow \prod_v H^1(K_v, E)[\phi]\}$, the kernel of the dashed arrow in the diagram. The **Tate-Shafarevich group** is $\text{III}(E/K) = \ker\{H^1(K, E) \rightarrow \prod_v H^1(K_v, E)\}$. Note that $\text{III}(E/K)$ is independent of ϕ

A diagram chase gives a s.e.s

$$0 \rightarrow E'(K)/\phi E(K) \rightarrow Sel_\phi(E/K) \rightarrow \text{III}(E/K)[\phi] \rightarrow 0$$

Take $\phi = [n]$ and we get

$$0 \rightarrow E(K)/nE(K) \rightarrow Sel_n(E/K) \rightarrow \text{III}(E/K)[n] \rightarrow 0$$

We should compare this with the s.e.s of the n -Selmer group of number field K

$$0 \rightarrow \mathcal{O}_K^\times/(\mathcal{O}_K^\times)^n \xrightarrow{i} Sel_n(K) \xrightarrow{\Pi} Cl(K)[n] \rightarrow 0$$

Theorem 17.4. For $n \geq 2$, $Sel_n(E/K)$ is finite.

It follows immediately that $E(K)/nE(K)$ is finite, i.e. the weak Mordell-Wéil holds.

Proof. For any L/K finite Galois, we have the inflation restriction exact sequence

$$\begin{array}{ccccc} 0 & \longrightarrow & H^1(Gal(L/K), E(L)[n]) & \xrightarrow{inf} & H^1(K, E[n]) & \xrightarrow{res} & H^1(L, E[n]) \\ & & & & \uparrow & & \uparrow \\ & & & & Sel_n(E/K) & \longrightarrow & Sel_n(E/L) \end{array}$$

Note that $H^1(Gal(L/K), E(L)[n])$ is finite since both groups are finite. So $\ker(Sel_n(E/K) \rightarrow Sel_n(E/L))$ is finite. Therefore, to show the result, we can extend K . In particular, we can assume that $E[n] \subset E(K)$. Hence $E[n] \cong \mu_n \times \mu_n$ is a Γ_K -module with trivial action and hence $H^1(K, E[n]) \cong H^1(K, \mu_n) \times H^1(K, \mu_n) \cong \frac{K^\times}{(K^\times)^n} \times \frac{K^\times}{(K^\times)^n}$

Now let $S = \{\text{places of bad reduction for } E\} \cup \{\text{places dividing } n\infty\}$. We write $H_S^1(K, A) = \ker(H^1(K, A) \xrightarrow{\prod_{v \notin S} res_v} \prod_{v \notin S} H^1(K_v^{nr}, A))$. We should think of $H_S^1(K, A)$ as cocycle classes unramified outside S .

For $v \notin S$ we have a commutative diagram with exact rows

$$\begin{array}{ccccc} E(K_v) & \xrightarrow{[n]} & E(K_v) & \xrightarrow{\delta_v} & H^1(K_v, E[n]) \\ \downarrow & & \downarrow & & \downarrow \text{res} \\ E(K_v^{nr}) & \xrightarrow{[n]} & E(K_v^{nr}) & \xrightarrow{\delta_v^{nr}} & H^1(K_v^{nr}, E[n]) \end{array}$$

by taking the Galois cohomology of Γ_{K_v} and $\Gamma_{K_v^{nr}}$. Now $E(K_v^{nr}) \xrightarrow{[n]} E(K_v^{nr})$ is surjective which is shown in the proof of theorem 11.3 ($E[n]$ is unramified when E has good reduction). So $\delta_v^{nr} = 0$ and so $\text{res} \circ \delta_v = 0$ and $\text{im}(\delta_v) \subset \ker(H^1(K_v, E[n]) \rightarrow H^1(K_v^{nr}, E[n]))$. Thus for any $\alpha \in \text{Sel}_n(E/K)$, $\text{res}_v(\alpha) \in \text{im}(\delta_v)$ for all v implies that $\text{Sel}_n(E/K) \subset H_S^1(K, E[n])$.

As above, $H_S^1(K, E[n]) \cong H_S^1(K, \mu_n) \times H_S^1(K, \mu_n)$. We claim that $K(S, n) = \{x \in K^\times / (K^\times)^n \mid v_p(x) \equiv 0 \pmod n \forall p \notin S\} \cong H_S^1(K, \mu_n)$. This is essentially the same as lemma 17.1. By identifying $H^1(K, \mu_n) \cong K^\times / (K^\times)^n$ and $H^1(K_v^{nr}, \mu_n) \cong K_v^{nr \times} / (K_v^{nr \times})^n$ via Hilbert's 90, we need to show the sequence

$$0 \rightarrow K(S, n) \rightarrow K^\times / (K^\times)^n \rightarrow \prod_{v \notin S} K_v^{nr \times} / (K_v^{nr \times})^n$$

is exact, which is precisely the generalization of the sequence in the proof of lemma 17.1

Now we know $K(S, n)$ is finite by proposition 13.2. So $\text{Sel}_n(E/K)$ is finite. $\square$

Corollary 17.2. *Let $\phi : E \rightarrow E'$ an isogeny defined over K . Then $\text{Sel}_\phi(E/K)$ is finite.*

Remark 17.1. By Mordell-Wéil theorem, $E(K) \cong E(K)_{\text{tor}} \times \mathbb{Z}^{rk(E)} \implies E(K)/nE(K) \cong E(K)[n] \times (\mathbb{Z}/n\mathbb{Z})^{rk(E)} \implies \#|E(K)/nE(K)| = \#|E(K)[n]| \cdot n^{rk(E)}$. So computing Selmer groups can give us an upper bound of rank.

18 Descent by cyclic isogeny

Let K a perfect field, $\text{char}(K) \nmid n$ and an isogeny $\phi : E \rightarrow E'$ over K with degree n .

Assume that $E'[\widehat{\phi}] \cong \mathbb{Z}/n\mathbb{Z}$ generated by $T \in E'(K)$. Then $E'[\widehat{\phi}] \cong \mu_n$ as Γ_K -module with trivial action and by corollary 13.1, $\eta : E[\phi] \cong \text{Hom}(E'[\widehat{\phi}], \mu_n) \cong \mu_n$ by $P \mapsto e_\phi(P, T)$. We then have a composition

$$\alpha : E'(K) \xrightarrow{\delta} H^1(K, E[\phi]) \cong H^1(K, \mu_n) \cong K^\times / (K^\times)^n$$

from the l.e.s of $0 \rightarrow E[\phi] \rightarrow E \xrightarrow{\phi} E' \rightarrow 0$ and Hilbert's 90.

Now since $nT = 0$ and $T \in E'[\widehat{\phi}]$, as in the construction of Wéil pairing, there is $f \in K(E')^\times$ such that $\text{div}(f) = n(T) - n(\mathcal{O})$ and $g \in K(E)$ such that $g^n = \phi^* f$.

Theorem 18.1. $\alpha(P) \equiv f(P) \pmod{(K^\times)^n}$ for all $P \in E'(K) \setminus \{T, \mathcal{O}\}$

Proof. Let $Q \in \phi^{-1}(P)$. Note that $\delta(P)$ is represented by the cocycle $\sigma \mapsto \sigma(Q) - Q$ for $\sigma \in \Gamma_K$. We have $\eta(\sigma(Q) - Q) = e_\phi(\sigma(Q) - Q, T) = \frac{g(\sigma(Q) - Q + R)}{g(R)}$ for any $R \in E(K)$ where g has no pole or zero.

We can take $R = Q$. Then $\frac{g(\sigma(Q) - Q + R)}{g(R)} = \frac{g\sigma(Q)}{g(Q)} = \frac{\sigma g(Q)}{g(Q)} = \frac{\sigma \sqrt[n]{f(P)}}{\sqrt[n]{f(P)}}$ since $g \in K(E)$ and $g^n = f \circ \phi$.

But $K^\times / (K^\times)^n \xrightarrow{\sim} H^1(K, \mu_n)$ is given by $x \mapsto (\sigma \mapsto \frac{\sigma(\sqrt[n]{x})}{\sqrt[n]{x}})$. So $f(P)$ represents the same cocycle as $\delta(P)$ under the identification. So $f(P) = \alpha(P) \in K^\times / (K^\times)^n$ $\square$

18.1 Descent by 2-isogeny

Recall that we have 2-isogeny as in example 7. Let $E/K : y^2 = x(x^2 + ax + b)$ and $E'/K : y^2 : x(x^2 + a'x + b')$ with $a' = -2a$ and $b' = a^2 - 4b$. Then $\phi : E \rightarrow E'$ given by $(x, y) \mapsto ((\frac{y}{x})^2, \frac{y}{x^2}(x^2 - b))$ and $\hat{\phi} : E' \rightarrow E$ given by $(x, y) \mapsto (\frac{1}{4}y^2/x^2, \frac{y(x^2 - b')}{8x^2})$ are isogenies of degree 2. $E[\phi] = \langle (0, 0) \rangle \cong \mathbb{Z}/2\mathbb{Z}$ and $E'[\hat{\phi}] = \langle (0, 0) \rangle \cong \mathbb{Z}/2\mathbb{Z}$

Now for $x \in K(E')$ we have $\text{div}(x) = 2(0, 0) - 2(\mathcal{O})$ and $\phi^*(x) = y^2/x^2$. So we can take $f = x$ and $g = y/x$ with $T = (0, 0)$ in theorem 18.1. So we get an injection

$$\alpha_\phi : E'(K)/\phi E(K) \rightarrow K^\times/(K^\times)^2$$

sending $P = (x, y) \neq T, \mathcal{O}$, i.e. $x \neq 0$ to $f(P) = x \bmod (K^\times)^2$. One can also show that $\alpha_\phi(T) = \alpha_\phi((0, 0)) = b' \bmod (K^\times)^2$

Similarly, we have an injection $\alpha_{\hat{\phi}} : E(K)/\hat{\phi} E'(K) \rightarrow K^\times/(K^\times)^2$ given by $P = (x, y) \mapsto x \bmod (K^\times)^2$ if $x \neq 0$ and $(0, 0) \mapsto b \bmod (K^\times)^2$

Now take $K = \mathbb{Q}$. We have a commutative diagram

$$\begin{array}{ccc} E(\mathbb{Q})/\hat{\phi} E'(K) & \xrightarrow{\alpha_{\hat{\phi}}} & \mathbb{Q}^\times/(\mathbb{Q}^\times)^2 \\ \downarrow & & \downarrow \\ E(\mathbb{Q}_p)/\hat{\phi} E'(K) & \xrightarrow{\alpha_{\hat{\phi}, p}} & \mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2 \end{array}$$

with $\mathbb{Q}_p = \mathbb{R}$ with $p = \infty$. By identifying $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ with $H^1(\mathbb{Q}, E'[\hat{\phi}])$ and $\mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2$ with $H^1(\mathbb{Q}_p, E'[\hat{\phi}])$, we have $\text{Sel}_{\hat{\phi}}(E'/\mathbb{Q}) = \{d \in \mathbb{Q}^\times/(\mathbb{Q}^\times)^2 \mid d \in \text{im}(\alpha_{\hat{\phi}, p}) \forall \text{ place } p\}$ and in particular $\text{im}(\alpha_{\hat{\phi}}) \subset \text{Sel}_{\hat{\phi}}(E'/\mathbb{Q})$. Note that under this identification the restriction becomes the natural map $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2 \rightarrow \mathbb{Q}_p^\times/(\mathbb{Q}_p^\times)^2$, and we can take d square-free integers.

Proposition 18.1. (Descent by 2-isogeny) Given $E/\mathbb{Q} : y^2 = x(x^2 + ax + b)$ with $a, b \in \mathbb{Z}$

1. $2^{rkE} = \frac{|\text{im}\alpha_\phi||\text{im}\alpha_{\hat{\phi}}|}{4} \leq \frac{|\text{Sel}_{\hat{\phi}}(E'/\mathbb{Q})||\text{Sel}_\phi(E/\mathbb{Q})|}{4}$
2. $|\text{im}\alpha_{\phi, p}||\text{im}\alpha_{\hat{\phi}, p}| = 2$ if $p = \infty$, 8 if $p = 2$ and 4 otherwise
3. $\text{Sel}_{\hat{\phi}}(E'/\mathbb{Q}) \subset \{d \text{ square-free integers} \mid d|b\}$, i.e. elements in $\text{Sel}_{\hat{\phi}}(E'/\mathbb{Q})$ has the form of square-free factors of b
4. Let $d_1 d_2 = b$ with d_1, d_2 square-free. Consider the equation $w^2 = d_1 u^4 + a u^2 v^2 + d_2 v^4$. Then $d_1 \in \text{Sel}_{\hat{\phi}}(E'/\mathbb{Q})$ if the equation is soluble over $\mathbb{Q}_p \forall p \nmid 2bb'\infty$ and $d_1 \in \text{im}\alpha_{\hat{\phi}}$ if and only if the equation is soluble over $\mathbb{Q}$

Proof. (1) and (2) are left in the example sheet. For (3), since for $d \in \text{Sel}_{\hat{\phi}}(E'/\mathbb{Q})$, $d \in \text{im}(\alpha_{\hat{\phi}, p})$ for all p , it suffices to show that for $p \nmid b$ and $(x, y) \in E(\mathbb{Q}_p)$, $v_p(x) \equiv 0 \bmod 2$. If $v_p(x) > 0$, then $v_p(x^2 + ax + b) = 0$ since b is a unit. So $v_p(x) = v_p(x(x^2 + ax + b)) = v_p(y^2) = 2v_p(y)$

If $v_p(x) = 0$, it is trivial

If $v_p(x) < 0$, by lemma 11.3, $v_p(x) = -2s, v_p(y) = -3s$ for $s \geq 1$.

(4) is more or less just change of variables. For $F \in \{\mathbb{Q}, \mathbb{Q}_p, \mathbb{R}\}$ we claim that the equation is soluble over F if and only if $d_1 \bmod (F^\times)^2 \in \alpha_{\hat{\phi}}(E(F))$

Note that $d_1 \bmod (F^\times)^2 \in \alpha_{\hat{\phi}}(E(F))$ if and only there is $(x, y) \in E(F)$ with $x = d_1 t^2$ for some $t \in (F^\times)^2$. Then $y^2 = d_1 t^2 (d_1^2 t^4 + a d_1 t^2 + b) \implies (y/d_1 t)^2 = d_1 t^4 + a t^2 + b/d_1$ (which is just d_2)

Conversely, suppose $(u, v, w) \neq 0$ is a solution in F for the equation. Then $uv \neq 0$ since d_1, d_2 square-free. Then $P = (d_1 u^2/v^2, d_1 uv/v^2) \in E(F)$. and we have $\alpha_{\hat{\phi}}(P) = d_1 \pmod{(F^\times)^2}$

Hence we get that $d_1 \in \text{im} \alpha_{\hat{\phi}}$ if and only if the equation is soluble over $\mathbb{Q}$ and $d_1 \in \text{Sel}_{\hat{\phi}}(E'/\mathbb{Q})$ if the equation is soluble over $\mathbb{Q}_p$ for all p .

Moreover, for $p \nmid 2bb'\infty$, by (3) $v_p(d) \equiv 0 \pmod 2$ for $d \in \text{im}(\alpha_{\hat{\phi},p})$, we have $\text{im}(\alpha_{\hat{\phi},p}) \subset \mathbb{Z}_p^\times/(\mathbb{Z}_p^\times)^2 \cong \mathbb{Z}/2\mathbb{Z}$ since $p \neq 2$. Now by (2) we see that $\text{im}(\alpha_{\hat{\phi},p}) = \mathbb{Z}_p^\times/(\mathbb{Z}_p^\times)^2$. So the solubility is automatic. $\square$

Remark 18.1. Proposition 18.1 essentially tells us how to compute Selmer groups. Note that since $\alpha_{\hat{\phi}}((0,0)) = b$, we always have $b \in \text{im}(\alpha_{\hat{\phi}}) \subset \text{Sel}_{\hat{\phi}}(E'/\mathbb{Q})$ and $b' \in \text{im}(\alpha_{\phi})$

18.2 Exercise

Proposition 18.2. *Prove proposition 18.1 (1)(2)*

Proof. First since $\alpha_{\hat{\phi}}$ is injective we can compute $|E(\mathbb{Q})/\hat{\phi}E'(K)|$ instead. We need the following fact from group theory:

Lemma 18.1. *Suppose we have group homomorphisms $A \xrightarrow{f} B \xrightarrow{g} C$, then there is an exact sequence*

$$0 \rightarrow \ker f \rightarrow \ker gf \rightarrow \ker g \rightarrow \text{coker} f \rightarrow \text{coker} gf \rightarrow \text{coker} g \rightarrow 0$$

Assuming this, we apply it to $E(\mathbb{Q}) \xrightarrow{\phi} E'(\mathbb{Q}) \xrightarrow{\hat{\phi}} E(\mathbb{Q})$ with $\phi, \hat{\phi}$ the 2-isogenies. So we get

$$0 \rightarrow E(\mathbb{Q})[\phi] \rightarrow E(\mathbb{Q})[2] \rightarrow E'(\mathbb{Q})[\hat{\phi}] \rightarrow E'(\mathbb{Q})/\phi E(\mathbb{Q}) \rightarrow E(\mathbb{Q})/2E(\mathbb{Q}) \rightarrow E(\mathbb{Q})/\hat{\phi}E'(K) \rightarrow 0$$

Now note that $|E(\mathbb{Q})[\phi]| = |E'(\mathbb{Q})[\hat{\phi}]| = 2$, and also $|E(\mathbb{Q})/2E(\mathbb{Q})| = |E(\mathbb{Q})[2]| \times 2^{\text{rank} E}$ by the remark after Mordell-Wéil. Combine all this and we get

$$|E(\mathbb{Q})/\hat{\phi}E'(K)| \times |E'(\mathbb{Q})/\phi E(\mathbb{Q})| = 4 \cdot \frac{|E(\mathbb{Q})/2E(\mathbb{Q})|}{|E(\mathbb{Q})[2]|} = 4 \cdot 2^{\text{rank} E}$$

We can also replace $\mathbb{Q}$ with $\mathbb{Q}_p$. In this case by proposition 14.2, $\frac{|E(\mathbb{Q}_p)/2E(\mathbb{Q}_p)|}{|E(\mathbb{Q}_p)[2]|} = 4 \cdot |\mathbb{Z}_p/2\mathbb{Z}_p| = 8$ if $p = 2$ and 4 otherwise by Hensel's lemma.

For $\mathbb{R}$, by theorem 7.1 we have $E(\mathbb{R}) \cong \mathbb{R}/\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ if $\Delta(E) > 0$ and $E(\mathbb{R}) \cong \mathbb{R}/\mathbb{Z}$ if $\Delta(E) < 0$, and we see $\frac{|E(\mathbb{R})/2E(\mathbb{R})|}{|E(\mathbb{R})[2]|} = 2$ in both cases. $\square$

Proposition 18.3. *Let d be a positive squarefree integer and let $E_d : y^2 = x^3 - d^2x$. Show that $E_d(\mathbb{Q})_{\text{tor}} = \{\mathcal{O}_E, (0,0), (\pm d, 0)\}$. Deduce that d is a congruent number if and only if $\text{rank}(E_d(\mathbb{Q})) > 0$*

Proof. Let $p \nmid 2d$ a prime such that $p \equiv 3 \pmod 4$. Recall that $|\overline{E_d}(\mathbb{F}_p)| = p + 1 + \sum_{x=0}^{p-1} (\frac{x^3 - d^2x}{p})$. Now $f(x) = x^3 - d^2x$ is an odd function and so $(\frac{f(x)}{p}) = (\frac{-f(-x)}{p}) = (\frac{-1}{p})(\frac{f(-x)}{p}) = -(\frac{f(-x)}{p})$. So the sum gets cancelled out and hence $|\overline{E_d}(\mathbb{F}_p)| = p + 1$.

Suppose ℓ an odd prime divides $|E_d(\mathbb{Q})_{\text{tor}}|$. For any prime $p \nmid 2d\ell$ with $p \equiv 3 \pmod 4$ we have $\ell \mid |\overline{E_d}(\mathbb{F}_p)| = p + 1$ as in proposition 12.4. So $p \equiv -1 \pmod \ell$. Then some residue class modulo $4d\ell$ can't contain primes, which violates Dirichlet's progression theorem. Similarly we show $8 \nmid |E_d(\mathbb{Q})_{\text{tor}}|$. Now since the 4 points in the statement are 2-torsion, we are done.

Now by proposition and remark 1.1, d is congruent if and only if E_d has a point $P = (x, y)$ with $y \neq 0$. Now by first part P is not torsion, which implies that $\text{rank}(E_d) > 0$ $\square$

Proposition 18.4. *Prove that 2 is not a congruent number. Find the rank of $y^2 = x^3 - p^2x$ for p a prime with $p \equiv 3 \pmod 8$.*

Proof. By proposition it suffices to show that $E : y^2 = x^3 - 4x$ has rank 0. Now we consider the 2-isogeny $\phi, \hat{\phi}$ and $E' : y^2 = x^3 + 16x$. Now $b = -4 \in \text{im}\alpha_{\hat{\phi}} \implies -1 \in \text{im}\alpha_{\hat{\phi},\infty} \implies \text{im}\alpha_{\phi,\infty} = 0$ by proposition 18.1 (2). So $\text{Sel}_{\phi}(E/\mathbb{Q}) \subset \{1, 2\}$. Also $\text{Sel}_{\hat{\phi}}(E'/\mathbb{Q}) \subset \{\pm 1, \pm 2\}$. Now we consider the equation $w^2 = 2u^4 + 8v^4$. By considering the valuation at 2 modulo 4 this is not soluble over $\mathbb{Q}$. So $2 \notin \text{Sel}_{\phi}(E/\mathbb{Q})$ and hence $\text{Sel}_{\phi}(E/\mathbb{Q}) = \{1\}$. Now by proposition 18.1 we have $\text{rank}(E) = 0$.

For $E : y^2 = x^3 - p^2x$ for p a prime with $p \equiv 3 \pmod{8}$, $E' = x^3 + 4p^2x$. Then $b = -p^2 \in \text{im}\alpha_{\hat{\phi}} \implies -1 \in \text{im}\alpha_{\hat{\phi},\infty} \implies \text{im}\alpha_{\phi,\infty} = 0$ by proposition 18.1 (2). So $\text{Sel}_{\phi}(E/\mathbb{Q}) \subset \{1, 2, p, 2p\}$ and $\text{Sel}_{\hat{\phi}}(E'/\mathbb{Q}) \subset \{\pm 1, \pm p\}$. Also note that $P = (p, 0) \in E(\mathbb{Q})$ and so $p \in \text{im}\alpha_{\hat{\phi}}$. So $\text{Sel}_{\hat{\phi}}(E'/\mathbb{Q}) = \{\pm 1, \pm p\}$. Since $p \equiv 3 \pmod{8}$, $-1 \notin (\mathbb{Q}_p)^2$. Hence -1 and p form a subgroup of order 4 in $\mathbb{Q}_p/\mathbb{Q}_p^2$. By proposition 18.1 (2), this implies $\text{im}\alpha_{\phi,p} = 0$ and hence $\text{Sel}_{\phi}(E/\mathbb{Q}) = 0$ since $2 \notin \mathbb{Q}_p^2$ since $p \equiv 3 \pmod{8}$. Hence $\text{rank}(E) = 0$ $\square$

Proposition 18.5. *Let $\omega(n)$ be the number of distinct prime factors of an integer n . Show that if $E/\mathbb{Q}$ is an elliptic curve given by $y^2 = x^3 + ax^2 + bx$, then*

$$\text{rank}(E/\mathbb{Q}) \leq \omega(b) + \omega(a^2 - 4b)$$

By considering real solubility, show that the inequality is strict.

Proof. We can use 2-isogeny. Since $\text{Sel}_{\phi}(E/\mathbb{Q})$ is a subset of squarefree integers dividing $b' = a^2 - 4b$, let $p_1, p_2, \dots, p_k$ be the distinct prime factors of $a^2 - b$ with $k = \omega(a^2 - b)$. Then the squarefree integers dividing $b' = a^2 - 4b$ are exactly elements in the set $S = \{(-1)^{a_0} \prod p_i^{a_i} \mid a_i \in \{0, 1\}\}$. Since $|S| = 2^{k+1} = 2^{1+\omega(a^2-4b)}$, $|\text{Sel}_{\phi}(E/\mathbb{Q})| \leq 2^{1+\omega(a^2-4b)}$. Similarly for $|\text{Sel}_{\hat{\phi}}(E'/\mathbb{Q})|$. Hence by proposition 18.1 (1) $\text{rank}(E/\mathbb{Q}) \leq \omega(b) + \omega(a^2 - 4b)$.

The equality holds if and only if $\text{Sel}_{\phi}(E/\mathbb{Q}) = S$ and same for $|\text{Sel}_{\hat{\phi}}(E'/\mathbb{Q})|$. But if $-1 \in \text{Sel}_{\phi}(E/\mathbb{Q})$, then $-1 \in \text{im}\alpha_{\phi,\infty} \implies \text{im}\alpha_{\hat{\phi},\infty} = 0$ by proposition 18.1 (2). So $\text{Sel}_{\hat{\phi}}(E'/\mathbb{Q})$ only contains positive integers and hence can't be the whole squarefree integers dividing b . So the inequality is strict. $\square$

19 Afterthoughts

It is truly a miracle to cover the main body of AEC in an 8-week 24-lecture-hour course without skipping or even sketching the proof of any major theorem or proposition (except for Riemann-Roch and Kummer's theory), even though I do think the proofs related to quadratic forms should be sketched at least. In contrast, they frequently appear as bookwork in exams.

Speaking of the exam, this is some typical 3-hour Cambridge-style exam: 4 questions out of 5, each consists of a heavy bookwork part and a tricky application part. I am very divided on the bookwork part: Some of them are meaningful, like recovering Weak Mordel-Weil assuming Kummer's theorem or even stuff on formal groups; Some are just some tedious tricks, like proving the canonical height is a quadratic form. Overall I think it is a pretty standard and intense exam even though I didn't do well in it.

What kind of bothers me is the order of teaching. It feels weird to put dual isogenies and Weil pairing near the end of this course. Overall this is still a legitimate course on elliptic curves, although somehow my interest in elliptic curves declines after all this...